

OPTIMAL TRADING IN COMMODITY MARKETS:

A Stochastic Optimal Control Approach under Transaction Costs, Predictable
Returns, and Stochastic Volatility

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Abstract

This thesis studies an optimal trading problem for commodities under predictable returns, transaction costs, and stochastic volatility. Extending the work of Chan, Sircar, and Zimbidis in equities [5], I adapt the model to a commodity setting by introducing price dynamics via the two-factor Schwartz model. This modification makes the associated Hamilton–Jacobi–Bellman equation nonlinear, so the problem is no longer tractable in closed form.

I formulate the problem in a stochastic optimal control framework and reduce the HJB equation through a quadratic ansatz in the inventory variable. The resulting system is solved numerically to converge to the viscosity solution of the HJB using a finite-difference scheme with backward Euler time stepping and upwind discretization. I then use Monte Carlo simulation to evaluate the performance of the strategy. In the empirical implementation, I calibrate a reduced one-factor stochastic-volatility model using WTI crude oil market data and the OVX volatility index. The numerical results indicate that stochastic volatility materially affects the optimal trading strategy. Compared to a constant-volatility benchmark, the stochastic-volatility strategy delivers higher expected profit and loss, but also a wider distribution of outcomes with positive skewness.

The code and data may be accessed in GitHub: https://github.com/aidenkaufman/Senior_Thesis/tree/main.

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Chapter 1

Introduction

Classical portfolio theory studies how an investor should allocate wealth over time in the presence of uncertainty. In continuous time, this question is often formulated as a stochastic optimal control problem: the environment evolves randomly, the investor makes decisions to interact with the environment, and the goal is to optimize an expected objective. Modern financial markets are shaped by uncertain environments such as trading frictions, predictable signals, and time-varying volatility. These features substantially change the form of the optimal policy for an investor.

This thesis studies an optimal trading problem in a commodity setting, extending the recent work on dynamic portfolio choice under return predictability, transaction costs, and stochastic volatility by Chan, Sircar, and Zimbidis [5]. They expand the framework of Gârleanu–Pedersen [9] by incorporating multiscale stochastic volatility into an optimal trading problem with market frictions. Here, I adapt this style of problem to commodity markets, where the underlying dynamics are tied directly to physical assets. This leads to a high-dimensional, nonlinear Hamilton–Jacobi–Bellman equation, which must generally be solved via numerical methods. Accordingly, this thesis develops the stochastic control formulation, derives the associated HJB equation, and studies its numerical approximation using finite-difference methods.

1.1 Motivation

Portfolio theory is one of the richest fields of financial mathematics research. In Merton's original portfolio problem, an investor must choose how to allocate their wealth between stocks and a risk-free asset so as to maximize expected utility. The problem is elegant because it converts an economic question into a stochastic control problem. This perspective has become foundational not only in asset allocation, but also in execution, hedging, and risk management.

Much of the classical literature, however, is written with equities in mind. This research, though, is especially important for commodities. Commodity markets raise additional modeling questions. Unlike stocks, commodities are tied to physical production, storage, transportation, seasonality, and consumption. Furthermore, commodity futures were created to form a symbiotic relationship. Producers and consumers could both guarantee a price for the product in advance to reduce uncertainty. A farmer, airline, refinery, or industrial user does not enter these markets merely to speculate, but to manage risk tied to an underlying physical good.

Today's commodity markets have become deeply financialized. Speculators affect prices and liquidity, driving markets away from their original purpose. As a result, commodity trading sits at the intersection of real economic fundamentals and financial market microstructure. This makes commodities a natural environment in which to study optimal trading with frictions.

Thus, the goal here is not simply to transport an equity model into a new asset class. It provides a tool to inform producers and consumers within a modern trading environment. Furthermore, the commodity dynamics makes the associated HJB equation less tractable analytically, which creates a need for careful numerical analysis. This project is then motivated both by financial utility and by mathematical complexity.

1.2 Background

This thesis builds on the paper *Optimal Trading under Instantaneous and Persistent Price Impact, Predictable Returns and Multiscale Stochastic Volatility* by Chan, Sircar, and Zimbidis [5].

Their framework extends the dynamic trading literature by incorporating four modeling choices that are especially relevant in practice.

First, the model allows for **return predictability**. Traders do not act in a vacuum; they use signals, factors, and forecasting models to estimate future returns. This is often represented through a mean-reverting signal process that influences the drift of the asset price.

Second, the model includes **transaction costs**. Adjusting a portfolio is not free. A model might suggest an immediate jump to a target position, but actual execution incurs costs such as bid–ask spread, commissions, and other temporary trading frictions. These costs discourage excessive trading.

Third, the model accounts for **price impact**. Large trades can move the market itself, so execution changes not only the investor’s position but also the price at which subsequent trading occurs.

Fourth, the model incorporates **stochastic volatility**. Financial volatility is not constant over time; it fluctuates between calm and volatile market regimes and often evolves on multiple time scales.

The mathematical framework for this thesis is stochastic optimal control. This type of problem is specified by a state process X , a control process α , and an objective function J . The state describes the evolving system, the control represents the agent’s actions, and the objective quantifies performance. The value function is then defined by

$$V := \sup_{\alpha \in \mathcal{A}} J(\alpha),$$

or by an infimum in a minimization problem. The central goal is to find the value function and identify an optimal control.

Dynamic programming leads to a Hamilton–Jacobi–Bellman equation for the value function. This PDE can sometimes be solved explicitly, but the HJB often becomes nonlinear and closed-form solutions are unavailable, and so we must turn to numerical analysis. That is the setting

of this thesis.

1.3 Related Work

Dynamic trading under market frictions is classically studied through linear–quadratic control formulations. In optimal execution, Almgren–Chriss [1] models trading as choosing a rate to balance execution costs against price-risk exposure, with instantaneous and persistent price impact terms leading to optimal liquidation moves. Others model persistent or transient impact through order-book resilience, where the execution cost depends on the recovery of supply and demand over time, as in Obizhaeva–Wang [13]. Gârleanu–Pedersen [9] derive closed-form dynamic portfolio rules with predictable returns and quadratic trading costs, showing that optimal trading amounts to partial adjustment toward an “aim” portfolio that looks forward to future targets implied by the signal.

Recent work integrates stochastic volatility into these trading models. Heston-type [10] specifications provide a canonical one-factor stochastic volatility structure in which variance is itself stochastic and can be interpreted as an instantaneous variance state. Multiscale stochastic volatility models treat slow and fast volatility factors through regular and singular perturbations of the HJB equation. Fouque–Sircar–Zariphopoulou [8] apply these ideas in portfolio optimization, and Chan–Sircar–Zimbidis [5] extend the predictable-return trading problem with both instantaneous and persistent transaction costs by deriving multiscale volatility corrections around a constant-volatility benchmark and evaluating the resulting strategies numerically. In contrast, the present thesis adapts the framework to a commodity setting by imposing Schwartz two-factor commodity price dynamics [14], and it uses a reduced one-factor stochastic volatility specification calibrated from OVX for numerical simulation.

On the mathematical and numerical side, the commodity dynamics makes the associated HJB equation nonlinear and generally not solvable in closed form. I apply a quadratic ansatz in inventory to obtain a coupled coefficient PDE system and then solve this system numerically via

a monotone finite-difference scheme. Convergence is justified at the viscosity-solution level. The viscosity framework of Crandall–Ishii–Lions [7] provides the appropriate notion of solution for nonlinear HJB equations, and the Barles–Souganidis theorem [3] gives convergence of monotone, stable, consistent schemes under a comparison principle. This places the numerical analysis of the thesis within the standard viscosity-solution approach to fully nonlinear control PDEs, while the empirical calibration and Monte Carlo evaluation connect the model back to observable data.

1.4 Organization of the Thesis

Chapter two reviews the stochastic optimal control framework and the theoretical results needed to formulate the problem. Chapter three develops the numerical methods used to approximate the resulting HJB equation, with particular attention to monotonicity, stability, and convergence. Chapter four introduces the commodity trading model, derives the associated HJB equation, and presents the numerical results. Chapter five concludes and provides next steps.

Chapter 2

Stochastic Optimal Control

In this section, we review the relevant background in the field of stochastic optimal control to understand the problem. I draw this section heavily from unpublished course notes by Emma Hubert [11]. Stochastic optimal control problems concern how an agent that controls a system with evolving stochastic dynamics can optimize some criterion.

2.1 Stochastic Processes

We should first understand the mathematical definitions behind stochastic processes that allow us to formulate such a problem.

2.1.1 Probability Definitions

Stochastic processes are collections of random variables, so we should start by defining a random variable. First, note that we call a pair $(S, \mathcal{F})$, consisting of an arbitrary set S with a σ -algebra $\mathcal{F}$ on it, a measurable space. If μ is a measure defined on the pair, then we call $(S, \mathcal{F}, \mu)$ a measure space (or a probability space if the measure is the probability measure $\mathbb{P}$).

Definition 2.1.1 (Random Variable). *A function $f : (\Omega, \mathcal{F}) \rightarrow (S, \mathcal{S})$ is called a measurable function if for any $A \in \mathcal{S}$, the preimage $f^{-1}(A) := \{x \in \Omega : f(x) \in A\}$ is also measurable. A*

random variable is a measurable function $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (S, \mathcal{S})$.

While a single random variable provides a static image of uncertainty, the systems we aim to control are dynamic. Their state evolves, and uncertainty unfolds continuously throughout time. To model this, we must extend our static definition into a collection of random variables. Let us define $\mathbb{T}$ as a subset of times within $[0, \infty) \subset \mathbb{R}$.

Definition 2.1.2 (Stochastic Process). *Let E be a subspace of a finite-dimensional space, and denote by $\mathcal{B}(E)$ the Borel σ -algebra on E . An E -valued stochastic process $X = (X_t)_{t \in \mathbb{T}}$ is a collection of E -valued random variables indexed by time on $[0, \infty)$. In other words, X is the following map:*

$$X : (t, \omega) \in \mathbb{T} \times \Omega \rightarrow X_t(\omega) \in E.$$

The process X is said to be measurable if it is measurable from $(\mathbb{T} \times \Omega, \mathcal{B}(\mathbb{T})) \otimes \mathcal{F}$ to $(E, \mathcal{B}(E))$.

We must now define what the controlling agent does actually know at any given moment. An agent cannot see the future, and so their decisions at time t must rely strictly on the information revealed up to time t . To mathematically model the accumulation of information, we introduce the concept of a filtration.

Definition 2.1.3 (Filtration). *A filtration on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is an increasing family $\mathbb{F} = (\mathcal{F}_t)_{t \in \mathbb{T}}$ of σ -fields of $\mathcal{F}$. A filtration satisfies the usual conditions if it is right-continuous such that $\mathcal{F}_{t+} := \bigcap_{s>t} \mathcal{F}_s = \mathcal{F}_t$ for all $t \in \mathbb{T}$ and if it is complete such that $\mathcal{F}_0$ contains the negligible sets of $\sigma(\bigcup_{t \in \mathbb{T}} \mathcal{F}_t)$.*

We will also typically assume that the process X is $\mathbb{F}$ -adapted, meaning the mapping $\omega \rightarrow X_t(\omega)$ is $\mathcal{F}_t$ -measurable for all $t \in \mathbb{T}$, or that it is $\mathbb{F}$ -progressively measurable, meaning the mapping $(s, \omega) \rightarrow X_s(\omega)$ is measurable on $[0, t] \times \Omega$ equipped with the product σ -field $\mathcal{B}([0, t]) \otimes \mathcal{F}_t$ for all $t \in \mathbb{T}$.

The adaptedness of a process ensures our agent only acts on available information. In many problems, we often want to make decisions based on when specific events occur. Because the

state is stochastic, the exact time this event occurs is also random. We must formalize random times that respect our filtration.

Definition 2.1.4 (Stopping Time). *A $[0, \infty]$ -valued random variable τ on $(\Omega, \mathcal{F}, \mathbb{P}, \mathbb{F})$ is called a random time. The random time τ is called an $\mathbb{F}$ -stopping time if for all $t \in \mathbb{T}$:*

$$\{\tau \leq t\} := \{\omega \in \Omega : \tau(\omega) \leq t\} \in \mathcal{F}_t.$$

With the infrastructure for stochastic processes established, we are almost equipped to formulate the optimal control problem. To evaluate how a function of our stochastic system changes over time, though, classical calculus is insufficient. The trajectories of the stochastic processes we deal with (namely, Brownian motion) are often continuous but nowhere differentiable. To operate on these paths, we require a new theorem based on stochastic calculus.

2.1.2 Stochastic Calculus

We now introduce the basic objects of stochastic calculus that make this possible. The filtered probability space $(\Omega, \mathcal{F}, \mathbb{P})$ carries with it a one-dimensional $\mathbb{F}$ -Brownian motion $W = (W_t)_{t \in \mathbb{T}}$.

Definition 2.1.5 (Brownian Motion). *A one-dimensional $\mathbb{F}$ -Brownian motion $W = (W_t)_{t \in \mathbb{T}}$ on $(\Omega, \mathcal{F}, \mathbb{P})$ is a continuous $\mathbb{F}$ -adapted process such that:*

- $W_0 = 0$ almost surely;
- for every $0 \leq s < t$, the increment $W_t - W_s$ is independent of $\mathcal{F}_s$;
- for every $0 \leq s < t$, the increment $W_t - W_s$ is normally distributed with mean 0 and variance $t - s$.

Brownian motion is the fundamental noise process in continuous-time finance and control. Its paths are continuous but nowhere differentiable in the classical sense. Brownian motions are a subset of the broader category of martingales.

Definition 2.1.6 (Martingale). *An $\mathbb{R}$ -valued process $M = (M_t)_{t \in \mathbb{T}}$ is called an $(\mathbb{F}, \mathbb{P})$ -martingale if it is $\mathbb{F}$ -adapted, integrable for every $t \in \mathbb{T}$, and satisfies*

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s \quad \text{almost surely for all } 0 \leq s \leq t.$$

Intuitively, a martingale says that your expected value later given the information you have now is your value now. If you flip a coin and gain a dollar for heads and lose a dollar for tails, your expected wealth later is the same as your wealth right now since the probabilities and values are symmetric. In order to differentiate with respect to these rough processes, we must also measure how the path fluctuates over very small time intervals, and this is captured by its quadratic variation.

Definition 2.1.7 (Quadratic Variation). *Let $X = (X_t)_{t \in \mathbb{T}}$ be a continuous process. If the following limit exists in probability, we call it the quadratic variation of X on $[0, t]$:*

$$[X]_t := \lim_{|\pi| \rightarrow 0} \sum_{i=0}^{n-1} (X_{t_{i+1}} - X_{t_i})^2,$$

where $\pi = \{0 = t_0 < t_1 < \dots < t_n = t\}$ is a partition of $[0, t]$ and $|\pi| := \max_i |t_{i+1} - t_i|$.

For smooth or finite-variation paths, this quantity is zero. Brownian motions behave differently, though. Their squares accumulate over time rather than disappearing.

Theorem 2.1.8 (Quadratic Variation of a Continuous Martingale). *If M is a continuous square-integrable martingale, then there exists an increasing process $\langle M \rangle = (\langle M \rangle_t)_{t \in \mathbb{T}}$ with $\langle M \rangle_0 = 0$ such that*

$$M_t^2 - \langle M \rangle_t$$

is a martingale. The process $\langle M \rangle$ is called the predictable quadratic variation of M . For continuous martingales, it coincides with the quadratic variation $[M]$.

Corollary 2.1.9 (Quadratic Variation of Brownian Motion). *If W is a one-dimensional Brow-*

nian motion, then

$$\langle W \rangle_t = t \quad \text{for all } t \in \mathbb{T}.$$

This is the crucial feature that distinguishes Brownian motion from an ordinary differentiable path. Because the squared increments of Brownian motion accumulate at rate dt , the second-order term in a Taylor expansion does not disappear in the limit. This is the source of the extra term in Itô's formula.

We now define integration with respect to Brownian motion. The construction begins with simple adapted processes and is then extended by continuity.

Definition 2.1.10 (Simple Adapted Process). *A process $\phi = (\phi_t)_{t \in \mathbb{T}}$ is called simple and adapted if it can be written in the form*

$$\phi_t = \sum_{i=0}^{n-1} \xi_i \mathbf{1}_{(t_i, t_{i+1}]}(t),$$

where $0 = t_0 < t_1 < \dots < t_n$, and each ξ_i is $\mathcal{F}_{t_i}$ -measurable.

Definition 2.1.11 (Stochastic Integral for Simple Processes). *Let ϕ be a simple adapted process as above. Its stochastic integral with respect to Brownian motion is defined by*

$$\int_0^t \phi_s dW_s := \sum_{i=0}^{n-1} \xi_i (W_{t_{i+1} \wedge t} - W_{t_i \wedge t}).$$

The requirement that ξ_i be $\mathcal{F}_{t_i}$ -measurable ensures that the integrand depends only on information available at the beginning of each interval. The following theorem takes us from simple processes to progressively measurable processes.

Theorem 2.1.12 (Itô Isometry). *Let ϕ be a progressively measurable process such that*

$$\mathbb{E} \left[\int_0^T |\phi_s|^2 ds \right] < \infty.$$

Then the stochastic integral $\int_0^t \phi_s dW_s$ is well-defined, is a square-integrable martingale, and

satisfies

$$\mathbb{E} \left[\left(\int_0^t \phi_s \, dW_s \right)^2 \right] = \mathbb{E} \left[\int_0^t \phi_s^2 \, ds \right]$$

for all $t \in \mathbb{T}$.

Itô isometry is the fundamental tool behind the construction of the stochastic integral. It shows that integration against Brownian motion behaves like an isometry from the space of square-integrable adapted processes into the space of square-integrable random variables.

Next, we utilize stochastic integration to define Itô processes. The stochastic processes that define our state variables in this thesis are generally of Itô type.

Definition 2.1.13 (Itô Process). *An $\mathbb{R}$ -valued process $X = (X_t)_{t \in \mathbb{T}}$ is called an Itô process if it admits the representation*

$$X_t = X_0 + \int_0^t b_s \, ds + \int_0^t \sigma_s \, dW_s,$$

where X_0 is $\mathcal{F}_0$ -measurable and b, σ are progressively measurable processes satisfying

$$\int_0^t |b_s| \, ds + \int_0^t |\sigma_s|^2 \, ds < \infty$$

almost surely for all $t \in \mathbb{T}$. Equivalently, in differential notation,

$$dX_t = b_t \, dt + \sigma_t \, dW_t.$$

The first term is the drift, which describes the deterministic component of motion, while the second term is the diffusion term, which captures randomness through Brownian motion.

Proposition 2.1.14 (Quadratic Variation of an Itô Process). *Let X and Y be Itô processes of the form*

$$dX_t = b_t^X \, dt + \sigma_t^X \, dW_t, \quad dY_t = b_t^Y \, dt + \sigma_t^Y \, dW_t.$$

Then their cross-variation is

$$\langle X, Y \rangle_t = \int_0^t \sigma_s^X \sigma_s^Y \, ds.$$

In particular,

$$\langle X \rangle_t = \int_0^t (\sigma_s^X)^2 ds.$$

This shows that only the diffusion term of an Itô process contributes to quadratic variation. The drift term has finite variation and therefore contributes nothing. Finally, this brings us to Itô's formula, which is a foundational tool that allows us to calculate the behavior of functions of stochastic processes.

Theorem 2.1.15 (Itô's Formula). *Let X be an Itô process satisfying*

$$dX_t = b_t dt + \sigma_t dW_t,$$

and let $f \in C^{1,2}(\mathbb{T} \times \mathbb{R})$. Then $(f(t, X_t))_{t \in \mathbb{T}}$ is again an Itô process, and

$$df(t, X_t) = \left(\frac{\partial f}{\partial t}(t, X_t) + b_t \frac{\partial f}{\partial x}(t, X_t) + \frac{1}{2} \sigma_t^2 \frac{\partial^2 f}{\partial x^2}(t, X_t) \right) dt + \sigma_t \frac{\partial f}{\partial x}(t, X_t) dW_t.$$

Equivalently, in integral form,

$$\begin{aligned} f(t, X_t) &= f(0, X_0) + \int_0^t \frac{\partial f}{\partial s}(s, X_s) ds + \int_0^t \frac{\partial f}{\partial x}(s, X_s) b_s ds \\ &\quad + \int_0^t \frac{\partial f}{\partial x}(s, X_s) \sigma_s dW_s + \frac{1}{2} \int_0^t \frac{\partial^2 f}{\partial x^2}(s, X_s) \sigma_s^2 ds. \end{aligned}$$

The formula may be understood as a stochastic version of the chain rule. To see why the additional second-derivative term appears, consider first the special case $X_t = W_t$ and a smooth function $f = f(x)$. Over a short interval, Taylor expansion gives

$$f(W_{t+\Delta t}) - f(W_t) \approx f'(W_t) \Delta W_t + \frac{1}{2} f''(W_t) (\Delta W_t)^2.$$

For an ordinary differentiable path, the quadratic term would be negligible. For Brownian motion, however, $(\Delta W_t)^2$ is of order Δt , and so it contributes in the limit. This is precisely the origin of

the extra term in Itô's formula. This also yields the special case of the stochastic version of the product rule.

Lemma 2.1.16 (Itô Product Rule). *If X and Y are Itô processes, then*

$$d(X_t Y_t) = X_t dY_t + Y_t dX_t + d\langle X, Y \rangle_t.$$

These results are the basic tools needed throughout stochastic optimal control. In particular, controlled diffusions are Itô processes, and Itô's formula is what allows us to analyze the evolution of candidate value functions and derive the Hamilton–Jacobi–Bellman equation.

2.2 Problem Formulation

A stochastic optimal control problem consists of three elements:

- **Stochastic:** A collection of random variables representing the evolution of a system over time or space. In other words, the system has randomness or uncertainty.
- **Optimal:** An objective function that evaluates some performance criterion, such as minimizing a cost function or maximizing a reward function. Because the system is stochastic, this typically involves optimizing the expected value of the function.
- **Control:** The sequence of actions or inputs applied to the system to influence its trajectory. This is usually formulated as a policy or feedback rule that dictates the best action to take based on the currently observed state.

To define these concepts mathematically, we must first fix a time horizon $\mathbb{T}$, which is either finite $\mathbb{T} = [0, T]$ or infinite $\mathbb{T} = [0, \infty)$. Then, we fix a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ with a one-dimensional Brownian motion W and state process X . We also have a control $\alpha = (\alpha_t)_{t \geq 0} \in \mathcal{A}$ where the set of admissible controls is defined as:

$$\mathcal{A}_t := \left\{ \mathbb{F}\text{-progressively measurable } \alpha_s \in A \subseteq \mathbb{R} \text{ a.s. such that } \mathbb{E} \left[\int_t^T |\alpha_s|^2 ds \right] < \infty \right\},$$

with $\mathcal{A} := \mathcal{A}_0$.

The control should be able to influence the state process. Thus, we let X follow a controlled diffusion process:

$$dX_t = b(t, X_t, \alpha_t) dt + \sigma(t, X_t, \alpha_t) dW_t, \quad (2.1)$$

with $X_0 = x \in \mathbb{R}$ and functions b and σ from $\mathbb{T} \times \mathbb{R} \times \mathcal{A}$ to $\mathbb{R}$. We assume the functions b and σ ensure existence and uniqueness for a strong solution to the SDE.

For the performance criterion, we will assume we are trying to maximize (respectively minimize) over all admissible controls some reward function $J : \mathbb{T} \times \mathbb{R} \times \mathcal{A} \rightarrow \mathbb{R}$. For the finite horizon case, we have three functions: the final reward $g : \mathbb{R} \rightarrow \mathbb{R}$, the running reward $f : \mathbb{T} \times \mathbb{R} \times \mathcal{A} \rightarrow \mathbb{R}$, and the discount factor $\kappa : \mathbb{T} \times \mathbb{R} \times \mathcal{A} \rightarrow \mathbb{R}$. The criterion J is defined as:

$$J(t, x, \alpha) := \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(T) g(X_T) + \int_t^T \mathcal{K}_{t,x,\alpha}(s) f(s, X_s, \alpha_s) ds \right],$$

where

$$\mathcal{K}_{t,x,\alpha}(s) := \exp \left(- \int_t^s \kappa(u, X_u, \alpha_u) du \right).$$

The value function $V : \mathbb{T} \times \mathbb{R} \rightarrow \mathbb{R}$ is the maximum (respectively minimum) of this criterion over all possible controls:

$$V(t, x) := \sup_{\alpha \in \mathcal{A}_t} J(t, x, \alpha), \quad (2.2)$$

and the goal of stochastic optimal control problems is to find the optimal control α^* such that $V(t, x) = J(t, x, \alpha^*)$.

The formulation above models a scenario with a strict, known deadline T . Many real-world systems, such as long-term portfolio management or population dynamics, do not possess a natural terminal date. In these scenarios, we formulate the problem on an infinite time horizon. Moving to an infinite horizon shifts our focus from reaching a terminal goal $g(x)$ to stabilizing and optimizing the system's long-term, steady-state behavior. Because integrating a running reward over an infinite horizon could trivially diverge to infinity, the strictly positive discount

factor κ becomes mathematically essential to ensure the convergence of the criterion J . Thus, for the infinite horizon case, our objective function is time-independent, possesses no terminal reward, and is given by:

$$J(x, \alpha) := \mathbb{E} \left[\int_0^\infty \mathcal{K}_{x, \alpha} f(X_s, \alpha_s) \, ds \right],$$

where

$$\mathcal{K}_{x, \alpha}(s) := \exp \left(- \int_0^s \kappa(X_u, \alpha_u) \, du \right),$$

and value function

$$V(x) := \sup_{\alpha \in \mathcal{A}} J(x, \alpha). \tag{2.3}$$

Our area of focus is thus on this triple (X, α, J) to solve for the value function and compute the optimal control.

2.3 Relevant Theorems

To approach solving such a system, we introduce a few theorems.

2.3.1 Dynamic Programming Principle

Define $X^{t, x, \alpha}$ as the strong solution starting from $x \in \mathbb{R}$ at time $t \in \mathbb{T}$ with control $\alpha \in \mathbb{R}$ to the SDE (2.1) as above. To ensure that such a solution exists and is unique, we need some assumptions on the functions:

Lemma 2.3.1. *If we assume there exists a deterministic constant $K \in \mathbb{R}$ such that for all $(t, x, a) \in \mathbb{T} \times \mathbb{R} \times \mathcal{A}$:*

$$|b(t, x, a) - b(t, y, a)| + |\sigma(t, x, a) - \sigma(t, y, a)| \leq K|x - y| \quad \forall y \in \mathbb{R},$$

$$|b(t, x, a)| + |\sigma(t, x, a)| \leq K(1 + |x| + |a|),$$

then we have that for all initial conditions $(t, x) \in \mathbb{T} \times \mathbb{R}$ and control processes $\alpha \in \mathcal{A}$, there exists

a unique strong solution $X^{t,x,\alpha}$ to the SDE (2.1). Additionally, for every $T > 0$, there exists a constant $C > 0$ such that:

$$\mathbb{E} \left[\sup_{t \leq s \leq T} |X_s^{t,x,\alpha}|^2 \right] \leq C(1 + |x|^2)e^{Ct} \quad \forall t \in [0, T]$$

We will assume that the criteria are met to satisfy this lemma. We will also assume that the functions f and κ are continuous, κ is bounded below by $\kappa_0 \in \mathbb{R}$, and f and g have quadratic growth for some constant $K \in \mathbb{R}$:

$$|f(t, x, a)| + |g(x)| \leq K(1 + |a| + |x|^2) \quad \forall (t, x, a) \in \mathbb{T} \times \mathbb{R} \times \mathcal{A}.$$

From here, we consider the classical version of the Dynamic Programming Principle.

Theorem 2.3.2 (Dynamic Programming Principle). *Let $(t, x) \in \mathbb{T} \times \mathbb{R}$. For value function V (2.2) and for any $[t, T]$ -valued stopping time τ , we have:*

$$V(t, x) = \sup_{\alpha \in \mathcal{A}_t} \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) V(\tau, X_\tau^{t,x,\alpha}) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}, \alpha_s) ds \right].$$

We will prove the strong version of this theorem.

Theorem 2.3.3 (Strong Dynamic Programming Principle). *Let $(t, x) \in [0, T] \times \mathbb{R}$ and let $\mathcal{T}_{t,T}$ be the set of $[t, T]$ -valued $\mathbb{F}$ -stopping times. Under our assumptions, we have:*

$$\begin{aligned} V(t, x) &= \sup_{\alpha \in \mathcal{A}_t} \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) V(\tau, X_\tau^{t,x,\alpha}) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}, \alpha_s) ds \right] \\ &= \sup_{\alpha \in \mathcal{A}_t} \inf_{\tau \in \mathcal{T}_{t,T}} \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) V(\tau, X_\tau^{t,x,\alpha}) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}, \alpha_s) ds \right]. \end{aligned}$$

Proof. Fix $(t, x) \in [0, T] \times \mathbb{R}$. First, we seek to show:

$$J(t, x, \alpha) = \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) J(\tau, X_\tau^{t,x,\alpha}) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}, \alpha_s) ds \right]$$

for a fixed admissible control $\alpha \in \mathcal{A}_t$ and stopping time $\tau \in \mathcal{T}_{t,T}$. Since $X^{t,x,\alpha}$ is a unique strong solution to its SDE, we have pathwise uniqueness such that for all $s \in [\tau, T]$ that $X_s^{t,x,\alpha} = X_s^{\tau, X_\tau^{t,x,\alpha}, \alpha}$. This implies:

$$\begin{aligned} \mathcal{K}_{t,x,\alpha}(s) &= \exp\left(-\int_t^s \kappa(u, X_u^{t,x,\alpha}, \alpha_u) du\right) \\ &= \exp\left(-\int_t^\tau \kappa(u, X_u^{t,x,\alpha}, \alpha_u) du\right) \exp\left(-\int_\tau^s \kappa(u, X_u^{t,x,\alpha}, \alpha_u) du\right) = \mathcal{K}_{t,x,\alpha}(\tau) \mathcal{K}_{\tau, X_\tau^{t,x,\alpha}, \alpha}(s). \end{aligned}$$

Then, by the tower property and Strong Markov property, we have:

$$\begin{aligned} J(t, x, \alpha) &= \mathbb{E} \left[\mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(T) g(X_T^{t,x,\alpha}) + \int_t^T \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}) ds \middle| \mathcal{F}_\tau \right] \right] \\ &= \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) \mathbb{E} \left[\mathcal{K}_{\tau, X_\tau^{t,x,\alpha}, \alpha}(T) g(X_T^{t,x,\alpha}) + \int_\tau^T \mathcal{K}_{\tau, X_\tau^{t,x,\alpha}, \alpha}(s) f(s, X_s^{t,x,\alpha}) ds \middle| \mathcal{F}_\tau \right] \right] \\ &\quad + \mathbb{E} \left[\int_t^\tau \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}) ds \right] \\ &= \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) J(\tau, X_\tau^{t,x,\alpha}) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}, \alpha_s) ds \right]. \end{aligned}$$

We will utilize this to prove two inequalities:

$$\begin{aligned} V(t, x) &\leq \sup_{\alpha \in \mathcal{A}_t} \inf_{\tau \in \mathcal{T}_{t,T}} \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) V(\tau, X_\tau^{t,x,\alpha}) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}, \alpha_s) ds \right], \\ V(t, x) &\geq \sup_{\alpha \in \mathcal{A}_t} \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) V(\tau, X_\tau^{t,x,\alpha}) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}, \alpha_s) ds \right]. \end{aligned}$$

Starting with the first inequality, for all $(s, y) \in [0, T] \times \mathbb{R}$, we have $J(s, y, \alpha) \leq V(s, y)$. Thus,

$$J(t, x, \alpha) \leq \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) V(\tau, X_\tau^{t,x,\alpha}) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}, \alpha_s) ds \right].$$

Since our stopping time τ is completely arbitrary, we have:

$$J(t, x, \alpha) \leq \inf_{\tau \in \mathcal{T}_{t,T}} \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) V(\tau, X_\tau^{t,x,\alpha}) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}, \alpha_s) ds \right],$$

and this is true for all admissible controls $\alpha \in \mathcal{A}_t$, we can take the supremum of both sides to get:

$$V(t, x) = \sup_{\alpha \in \mathcal{A}_t} J(t, x, \alpha) \leq \sup_{\alpha \in \mathcal{A}_t} \inf_{\tau \in \mathcal{T}_{t,T}} \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) V(\tau, X_\tau^{t,x,\alpha}) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}, \alpha_s) ds \right].$$

For the second inequality, the value function is defined as the supremum of J over all $\alpha \in \mathcal{A}_t$. We then have that for any $\epsilon > 0$, there exists an ϵ -optimal control $\alpha^\epsilon \in \mathcal{A}_\tau$ for $V(\tau, y)$ for all $y \in \mathbb{R}$ such that $V(\tau, y) - \epsilon \leq J(\tau, y, \alpha^\epsilon)$. Define, then, a new control

$$\hat{\alpha}_s := \begin{cases} \alpha_s & \text{if } s \in [t, \tau] \\ \alpha_s^\epsilon & \text{if } s \in [\tau, T] \end{cases}.$$

Under tedious measurability arguments via the measurable selection theorem, it can be shown that this control is progressively measurable and $\hat{\alpha} \in \mathcal{A}_t$, but we omit the proof here for simplicity. Thus, we can use the first equality we proved to see:

$$\begin{aligned} V(t, x) &\geq J(t, x, \hat{\alpha}) = \mathbb{E} \left[\mathcal{K}_{t,x,\hat{\alpha}}(\tau) J(\tau, X_\tau^{t,x,\hat{\alpha}}, \hat{\alpha}) + \int_t^\tau \mathcal{K}_{t,x,\hat{\alpha}}(s) f(s, X_s^{t,x,\hat{\alpha}}, \hat{\alpha}_s) ds \right] \\ &= \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) J(\tau, X_\tau^{t,x,\alpha}, \alpha^\epsilon) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}, \alpha_s) ds \right]. \end{aligned}$$

After time τ , we have that $J(\tau, X_\tau^{t,x,\alpha}, \alpha^\epsilon) \geq V(\tau, X_\tau^{t,x,\alpha}) - \epsilon$. Plugging this in above, we get:

$$V(t, x) \geq J(t, x, \hat{\alpha}) \geq \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) V(\tau, X_\tau^{t,x,\alpha}) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}, \alpha_s) ds \right] - \epsilon \mathbb{E} [\mathcal{K}_{t,x,\alpha}(\tau)].$$

From the same arbitrariness of α and τ as before, we can take the limit $\epsilon \rightarrow 0$ and get:

$$V(t, x) \geq \sup_{\alpha \in \mathcal{A}_t} \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) V(\tau, X_\tau^{t,x,\alpha}) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}, \alpha_s) ds \right].$$

Finally, we seek to combine these two inequalities. Notice first that:

$$\begin{aligned} & \sup_{\alpha \in \mathcal{A}_t} \inf_{\tau \in \mathcal{T}_{t,T}} \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) V(\tau, X_\tau^{t,x,\alpha}) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}, \alpha_s) ds \right] \\ & \leq \sup_{\alpha \in \mathcal{A}_t} \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) V(\tau, X_\tau^{t,x,\alpha}) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}, \alpha_s) ds \right], \end{aligned}$$

by definition of the supremum and infimum; however, we just proved that $V(t, x)$ is less than the first expression and greater than the second expression, which means that they all must be equal, yielding the desired result. $\square$

We also have the weak version of this theorem that we will omit the proof of.

Theorem 2.3.4 (Weak Dynamic Programming Principle). *Define the following functions for $(t, x) \in [0, T] \times \mathbb{R}$:*

$$V_*(t, x) := \liminf_{(t', x') \rightarrow (t, x)} V(t', x'), \quad V^*(t, x) := \limsup_{(t', x') \rightarrow (t, x)} V(t', x').$$

Assume that V (2.2) is locally bounded, fix $(t, x) \in [0, T] \times \mathbb{R}$, and let $(\tau^\alpha)_{\alpha \in \mathcal{A}_t}$ be a family of $[t, T]$ -valued $\mathbb{F}$ -stopping times. Then:

$$V(t, x) \leq \sup_{\alpha \in \mathcal{A}_t} \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau^\alpha) V^*(\tau^\alpha, X_{\tau^\alpha}^{t,x,\alpha}) + \int_t^{\tau^\alpha} \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}, \alpha_s) ds \right].$$

Assuming further that g is lower-semicontinuous and $X^{t,x,\alpha} \mathbb{1}_{[t, \tau^\alpha]}$ is $\mathbb{L}^\infty$ -bounded for all $\alpha \in \mathcal{A}_t$, then we have:

$$V(t, x) \geq \sup_{\alpha \in \mathcal{A}_t} \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau^\alpha) V_*(\tau^\alpha, X_{\tau^\alpha}^{t,x,\alpha}) + \int_t^{\tau^\alpha} \mathcal{K}_{t,x,\alpha}(s) f(s, X_s^{t,x,\alpha}, \alpha_s) ds \right].$$

If V is continuous, then $V = V_* = V^*$, and this theorem reduces to the classical Dynamic Programming Principle. This theorem makes no use of the measurability of V , and it is actually sufficient to prove the Hamilton–Jacobi–Bellman Equation that is the focus of the next subsection.

2.3.2 Hamilton–Jacobi–Bellman Equation

The Hamilton–Jacobi–Bellman equation is the infinitesimal counterpart to the Dynamic Programming equation that links the solution of a stochastic optimal control problem to the solution of a partial differential equation. Conceptually, if we take the time step in the DPP to be infinitesimally small (i.e., $\tau \rightarrow t$), we can apply Itô’s formula to unpack how the value function changes over a tiny fraction of time. For the derivation that appears below, we will make strong smoothness assumptions on the value function that are able to be relaxed in alternative proofs. We begin first by defining the Hamiltonian map $\mathcal{H}$.

Definition 2.3.5 (Hamiltonian Map). *For all $(t, x, y, z, \gamma) : \mathbb{T} \times \mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}$, the Hamiltonian Map $\mathcal{H}$ associated with the control problem is defined by:*

$$\mathcal{H}(t, x, y, z, \gamma) := \sup_{a \in A} \left\{ -\kappa(t, x, a)y + b(t, x, a)z + \frac{1}{2}\sigma^2(t, x, a)\gamma + f(t, x, a) \right\}.$$

The Hamilton–Jacobi–Bellman equation associated (as will be shown) with the control problem is:

$$-\partial_t v(t, x) - \mathcal{H}(t, x, v(t, x), \partial_x v(t, x), \partial_{xx}^2 v(t, x)) = 0, \quad (2.4)$$

with terminal condition $v(T, x) = g(x)$. Alternatively, we may rewrite this in terms of the generator $\mathcal{L}^a$:

$$-\sup_{a \in A} \{ \mathcal{L}^a v(t, x) - \kappa(t, x, a)v(t, x) + f(t, x, a) \} = 0,$$

where

$$\mathcal{L}^a \psi(t, x) := \partial_t \psi(t, x) + \partial_x \psi(t, x)b(t, x, a) + \frac{1}{2}\partial_{xx}^2 \psi(t, x)\sigma^2(t, x, a).$$

In order to connect the solution V of our value function for the control problem to the solution v of this partial differential equation, we have the following theorem. We will begin with the finite horizon problem.

Theorem 2.3.6. *Assume the value function V (2.2) is such that $V \in \mathcal{C}^{1,2}([0, T] \times \mathbb{R})$ and that the Hamiltonian map is continuous. Then, V is a solution to the HJB equation (2.4).*

Proof. Observe that V does satisfy the terminal condition $v(T, x) = g(x)$ since $V(T, x) = \sup_{\alpha \in \mathcal{A}_t} \mathbb{E}[g(X_T^{T,x,\alpha})] = g(x)$. From here, we first want to prove that V is a super solution to the HJB for all $a \in A$:

$$-\mathcal{L}^a V + \kappa^a V - f^a \geq 0.$$

Consider a constant control $\alpha_s = a \in A$ for all $s \in [0, T]$. Let $X = X^{t,x,\alpha}$ be the strong solution to its SDE (2.1) with this constant control and let τ be some stopping time that will be specified. For $s \in [t, \tau]$, we apply Itô's formula to $V(s, X_s)\mathcal{K}_{t,x,\alpha}(s)$ with

$$d(V(s, X_s)\mathcal{K}_{t,x,\alpha}(s)) = \mathcal{K}_{t,x,\alpha}(s)dV(s, X_s) - \kappa(s, X_s, \alpha)\mathcal{K}_{t,x,\alpha}(s)V(s, X_s)ds,$$

and

$$dV(t, X_t) = \mathcal{L}^a V(t, X_t)dt + \partial_x V \sigma(t, X_t, \alpha)dW_t,$$

since V satisfied the smoothness conditions, and this yields:

$$\begin{aligned} V(\tau, X_\tau)\mathcal{K}_{t,x,\alpha}(\tau) &= V(t, x) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s)[- \kappa^a V + \mathcal{L}^a V](s, X_s)ds \\ &\quad + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s)\partial_x V(s, X_s)\sigma(s, X_s, \alpha)dW_s. \end{aligned}$$

Define, then, τ to be the following stopping time:

$$\tau = (t + h) \wedge \inf\{s, |X_s - x| \geq 1\},$$

for $h > 0$. Notice that for h sufficiently close to 0, $\tau = t + h$. Taking expectations of our Itô's formula calculation and rearranging yields:

$$V(t, x) = \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) V(\tau, X_\tau) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s) [\kappa^a V - \mathcal{L}^a V](s, X_s) ds \right].$$

Additionally, by the Dynamic Programming Principle,

$$V(t, x) \geq \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau) V(\tau, X_\tau) + \int_t^\tau \mathcal{K}_{t,x,\alpha}(s) f(s, X_s, \alpha) ds \right].$$

Thus,

$$\mathbb{E} \left[\int_t^\tau \mathcal{K}_{t,x,\alpha}(s) [-\mathcal{L}^a V + \kappa^a V - f^a](s, X_s) ds \right] \geq 0.$$

Furthermore, based on our definition of the stopping time,

$$\lim_{h \rightarrow 0} \frac{1}{h} \mathbb{E} \left[\int_t^{t+h} \mathcal{K}_{t,x,\alpha}(s) [-\mathcal{L}^a V + \kappa^a V - f^a](s, X_s) ds \right] = [-\mathcal{L}^a V + \kappa^a V - f^a](t, X_t).$$

Thus, by the Dominated Convergence Theorem, for all $a \in A$:

$$-\mathcal{L}^a V + \kappa^a V - f^a \geq 0.$$

From here, we want to prove the reverse inequality. For this, we use the alternative definition of the HJB, seeking to show:

$$-\partial_t V(t, x) - \mathcal{H}(t, x, V(t, x), \partial_x V(t, x), \partial_{xx}^2 V(t, x)) \leq 0.$$

Assume, instead, that there exists a $(t_0, x_0) \in [0, T] \times \mathbb{R}$ such that:

$$-\partial_t V(t_0, x_0) - \mathcal{H}(t_0, x_0, V(t_0, x_0), \partial_x V(t_0, x_0), \partial_{xx}^2 V(t_0, x_0)) > 0.$$

Now, define $\psi : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ by

$$\psi(t, x) := V(t, x) + \epsilon((t - t_0)^2 + (x - x_0)^4)$$

for some $\epsilon > 0$. Observe that it is a smooth function since V is smooth. We have $\psi(t, x) \geq V(t, x)$ for all $(t, x) \in [0, T] \times \mathbb{R}$ and $\psi(t_0, x_0) = V(t_0, x_0)$. Additionally, all derivatives of $(\psi - V)$ are equal to zero at (t_0, x_0) . Because of this, we have:

$$-\partial_t \psi(t_0, x_0) - \mathcal{H}(t_0, x_0, \psi(t_0, x_0), \partial_x \psi(t_0, x_0), \partial_{xx}^2 \psi(t_0, x_0)) > 0.$$

Since the Hamiltonian map is continuous, there exists a sufficiently small constant $r > 0$ such that for $(t, x) \in \mathcal{N}_r := (t_0 - r, t_0 + r) \times [x_0 - r, x_0 + r]$:

$$h(t, x) := \partial_t \psi(t, x) + \mathcal{H}(t, x, \psi(t, x), \partial_x \psi(t, x), \partial_{xx}^2 \psi(t, x)) < 0,$$

or equivalently,

$$h(t, x) := \sup_{a \in A} \{[\mathcal{L}^a \psi - \kappa^a \psi + f^a](t, x)\} < 0.$$

Define a new function η where $\eta := -\max_{\partial \mathcal{N}_r} (V - \psi)$ which is > 0 by definition. Now, for some $\alpha \in \mathcal{A}_{t_0}$, define the stopping time τ^α such that

$$\tau^\alpha := \inf\{t > t_0 : (t, X_t^{t_0, x_0, \alpha}) \notin \mathcal{N}_r\}.$$

By pathwise continuity, $(\tau^\alpha, X_{\tau^\alpha}^{t_0, x_0, \alpha}) \in \partial \mathcal{N}_r$, and so

$$-\eta = \max_{\partial \mathcal{N}_r} (V - \psi) \geq (V - \psi)(\tau^\alpha, X_{\tau^\alpha}^{t_0, x_0, \alpha}).$$

Now, we take the expectation of Itô's formula on $\mathcal{K}_{t_0, x_0, \alpha}(t)\psi(t, X_t^{t_0, x_0, \alpha})$ between t_0 and τ^α :

$$\psi(t_0, x_0) = \mathbb{E} \left[\mathcal{K}_{t_0, x_0, \alpha}(\tau^\alpha) \psi(\tau^\alpha, X_{\tau^\alpha}^{t_0, x_0, \alpha}) - \int_{t_0}^{\tau^\alpha} \mathcal{K}_{t_0, x_0, \alpha}(t) [\mathcal{L}^\alpha \psi - \kappa^\alpha \psi](t, X_t^{t_0, x_0, \alpha}) dt \right],$$

since the expectation of the stochastic integral is 0 (because κ is bounded below and the choice of τ^α). Because $h(t, x) \geq [\mathcal{L}^a \psi - \kappa^a \psi + f^a](t, x)$ for all $(t, x) \in \mathcal{N}_r$ by definition of the supremum,

$$\psi(t_0, x_0) \geq \mathbb{E} \left[\mathcal{K}_{t_0, x_0, \alpha}(\tau^\alpha) \psi(\tau^\alpha, X_{\tau^\alpha}^{t_0, x_0, \alpha}) - \int_{t_0}^{\tau^\alpha} \mathcal{K}_{t_0, x_0, \alpha}(t) [f^a - h](t, X_t^{t_0, x_0, \alpha}) dt \right].$$

By the inequality for $-\eta$ (which we can use because for $t \in [t_0, \tau^\alpha]$ we have $(t, X_t^{t_0, x_0, \alpha}) \in \mathcal{N}_r$) and the fact that $h < 0$,

$$\psi(t_0, x_0) \geq \eta \mathbb{E} [\mathcal{K}_{t_0, x_0, \alpha}(\tau^\alpha)] + \mathbb{E} \left[\mathcal{K}_{t_0, x_0, \alpha}(\tau^\alpha) V(\tau^\alpha, X_{\tau^\alpha}^{t_0, x_0, \alpha}) + \int_{t_0}^{\tau^\alpha} \mathcal{K}_{t_0, x_0, \alpha}(t) f(t, X_t^{t_0, x_0, \alpha}, \alpha_t) dt \right].$$

Since we chose $\alpha \in \mathcal{A}_{t_0}$ arbitrarily and $\eta > 0$, we have:

$$V(t_0, x_0) = \psi(t_0, x_0) > \sup_{\alpha \in \mathcal{A}_{t_0}} \mathbb{E} \left[\mathcal{K}_{t_0, x_0, \alpha}(\tau^\alpha) V(\tau^\alpha, X_{\tau^\alpha}^{t_0, x_0, \alpha}) + \int_{t_0}^{\tau^\alpha} \mathcal{K}_{t_0, x_0, \alpha}(t) f(t, X_t^{t_0, x_0, \alpha}, \alpha_t) dt \right],$$

which violates the Dynamic Programming Principle, and so our assumption must have been incorrect. Thus, we have the other direction of the inequality. $\square$

We can also consider the infinite horizon counterpart to this theorem, given by the HJB equation for all $x \in \mathbb{R}$, smooth ψ , and constant $a \in A$:

$$-\sup_{a \in A} \{ \mathcal{L}^a v(x) - \kappa(x, a) v(x) + f(x, a) \} = 0, \quad (2.5)$$

where

$$\mathcal{L}^a \psi(x) := \partial_x \psi(x) b(x, a) + \frac{1}{2} \partial_{xx}^2 \psi(x) \sigma^2(x, a).$$

Theorem 2.3.7. *Assume the value function V (2.3) is such that $V \in \mathcal{C}^2(\mathbb{R})$ and that the Hamil-*

tonian map is continuous. Then, V is a solution to the HJB equation (2.5).

We omit the proof here because it follows the same pattern as the finite horizon version.

2.3.3 Verification Theorem

We now know that a sufficiently smooth value function V (2.2) is a solution to the HJB equation (2.4), so if we know V we can say that it solves the PDE. We really want to go the other direction, though. That is to say: we want that if v is a sufficiently smooth solution to the HJB equation, then it is a solution to the value function. Completing both directions establishes that they are one and the same problem. This brings us to the following theorem:

Theorem 2.3.8 (Verification Theorem). *Assume $w \in \mathcal{C}^0([0, T] \times \mathbb{R}) \cap \mathcal{C}^{1,2}([0, T) \times \mathbb{R})$ with quadratic growth in x .*

- *If w is a super-solution to the HJB equation and $w(T, x) \geq g(x)$, then $w(t, x) \geq V(t, x)$ for all $(t, x) \in [0, T] \times \mathbb{R}$.*
- *Assume there exists a measurable function $\hat{\alpha} : [0, T) \times \mathbb{R} \rightarrow A$ such that for all $(t, x) \in [0, T] \times \mathbb{R}$, X defined by*

$$dX_s = b(s, X_s, \hat{\alpha}(s, X_s))ds + \sigma(s, X_s, \hat{\alpha}(s, X_s))dW_s$$

has a strong solution for $s \in [t, T]$ and $X_t = x$, and such that

$$- \left\{ \mathcal{L}^{\hat{\alpha}(t, x)} w(t, x) - \kappa(t, x, \hat{\alpha}(t, x))w(t, x) + f(t, x, \hat{\alpha}(t, x)) \right\} \leq 0.$$

Furthermore, define the process α^ for all $s \in [t, T]$ by $\alpha_s^* := \hat{\alpha}(s, X_s)$ and assume it is in $\mathcal{A}_t$. If w also satisfies $w(T, x) \leq g(x)$, then $w(t, x) \leq J(t, x, \alpha^*)$.*

Proof. We will prove the first bullet point. Fix arbitrary $(t, x) \in [0, T] \times \mathbb{R}$ and $\alpha \in \mathcal{A}_t$. Consider a solution X to its SDE with control α between t and a stopping time τ^n for $n \in \mathbb{N}$. Applying

Itô's formula, we have:

$$\begin{aligned} \mathcal{K}_{t,x,\alpha}(\tau^n)w(\tau^n, X_{\tau^n}) &= w(t, x) + \int_t^{\tau^n} \mathcal{K}_{t,x,\alpha}(s)[- \kappa^{\alpha_s} w + \mathcal{L}^{\alpha_s} w](s, X_s) ds \\ &\quad + \int_t^{\tau^n} \mathcal{K}_{t,x,\alpha}(s) \partial_x w(s, X_s) \sigma(s, X_s, \alpha_s) dW_s. \end{aligned}$$

Define τ^n to be the following stopping time:

$$\tau^n = (T - \frac{1}{n}) \wedge \inf\{s, |X_s - x| \geq n\}.$$

Observe that $\tau^n \rightarrow T$ as $n \rightarrow \infty$. Taking the expectation of both sides of the previous equation yields,

$$w(t, x) = \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau^n)w(\tau^n, X_{\tau^n}) + \int_t^{\tau^n} \mathcal{K}_{t,x,\alpha}(s)[- \kappa^{\alpha_s} w + \mathcal{L}^{\alpha_s} w](s, X_s) ds \right],$$

since the expectation of the stochastic integral is 0. Because w is a super-solution to the HJB equation, this implies:

$$w(t, x) \geq \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(\tau^n)w(\tau^n, X_{\tau^n}) + \int_t^{\tau^n} \mathcal{K}_{t,x,\alpha}(s)f(s, X_s, \alpha_s) ds \right].$$

Taking the limit $n \rightarrow \infty$ of w yields:

$$\lim_{n \rightarrow \infty} w(\tau^n, X_{\tau^n}) = w(T, X_T) \geq g(X_T)$$

by continuity of w . Since w has quadratic growth, we can take the limit of both sides of the previous equation:

$$w(t, x) \geq \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(T)w(T, X_T) + \int_t^T \mathcal{K}_{t,x,\alpha}(s)f(s, X_s, \alpha_s) ds \right]$$

by Fatou's lemma. Since α is arbitrary, we may now take the supremum over all $\alpha \in \mathcal{A}_t$ to get:

$$w(t, x) \geq \sup_{\alpha \in \mathcal{A}_t} \mathbb{E} \left[\mathcal{K}_{t,x,\alpha}(T)w(T, X_T) + \int_t^T \mathcal{K}_{t,x,\alpha}(s)f(s, X_s, \alpha_s)ds \right] = V(t, x).$$

The second part of the theorem follows directly from these same steps with the opposite inequality. $\square$

By combining both points of the proof, we can obtain that $w(t, x) = V(t, x)$ for all $(t, x) \in [0, T] \times \mathbb{R}$. We again state here the infinite horizon version of the theorem.

Theorem 2.3.9 (Infinite Horizon Verification Theorem). *Assume $w \in C^2(\mathbb{R})$.*

- *If w is a super-solution to the HJB equation (2.5) such that for all $(x, \alpha) \in \mathbb{R} \times \mathcal{A}_0$:*

$$\liminf_{T \rightarrow \infty} \mathbb{E}[\mathcal{K}_{x,\alpha}(T)w(X_T^{x,\alpha})] \geq 0,$$

then $w(x) \geq V(x)$ for value function V (2.3) for all $x \in \mathbb{R}$.

- *Assume there exists a measurable function $\hat{\alpha} : \mathbb{R} \rightarrow A$ such that for all $x \in \mathbb{R}$, X defined by*

$$dX_s = b(X_s, \hat{\alpha}(X_s))ds + \sigma(X_s, \hat{\alpha}(X_s))dW_s$$

has a strong solution for $s \geq 0$ and $X_0 = x \in \mathbb{R}$, and such that

$$- \{ \mathcal{L}^{\hat{\alpha}(x)}w(x) - \kappa(x, \hat{\alpha}(x))w(x) + f(x, \hat{\alpha}(x)) \} \leq 0.$$

Furthermore, define the process α^ for all $s \geq 0$ by $\alpha_s^* := \hat{\alpha}(X_s)$ and assume it is in $\mathcal{A}_0$. If w also satisfies*

$$\limsup_{T \rightarrow \infty} \mathbb{E}[\mathcal{K}_{x,\alpha^*}(T)w(X_T^{x,\alpha^*})] \leq 0,$$

then $w(x) \leq J(x, \alpha^)$.*

Again, the proof follows the same steps as the finite version, so we omit the proof here.

With the Verification Theorem established, we have closed the knowledge we need of stochastic optimal control. We have shown not only that the value function of our control problem satisfies the HJB equation, but, crucially, that any sufficiently smooth solution to the HJB equation is the value function. This equivalence is incredibly powerful. It shifts our task from the nearly impossible challenge of searching over the infinite-dimensional space of all admissible controls $\mathcal{A}$ to the much more tractable task of solving a nonlinear, second-order partial differential equation. The smoothness restrictions are often too imposing in practice, specifically in regards to the problem in this thesis, which will lead us to the definition of viscosity solutions in the following section. Still, with these theorems having been proven, we have the core theoretical tools to approach the stochastic optimal control problem defined in this thesis.

Chapter 3

Numerical Analysis

As is the case in this thesis, stochastic optimal control problems do not always produce a closed-form solution. As such, we will need to turn to numerical methods for solving the Hamilton-Jacobi-Bellman Equation. In this section, we review the theoretical foundations of finite difference methods, focusing on the conditions required to guarantee that our numerical approximation converges to the true solution of the HJB equation.

3.1 Finite Difference Discretization

Let $\mathcal{L}v = 0$ represent a continuous PDE, where $\mathcal{L}$ is an infinitesimal generator associated with an underlying stochastic process. In the context of optimal control, $\mathcal{L}$ is typically a second-order elliptic or parabolic differential operator of the form $\mathcal{L}v = \partial_t v + b(t, x)\partial_x v + \frac{1}{2}\sigma^2(t, x)\partial_{xx}^2 v - rv$, encapsulating the drift, diffusion, and discounting of the system. To solve a PDE numerically, we must move from a continuous state space to a discrete one. We do this by defining a computational grid. For simplicity, let us consider a one-dimensional spatial domain $x \in [x_{\min}, x_{\max}]$ and a time domain $t \in [0, T]$. We partition the state space into N intervals of length $\Delta x = \frac{x_{\max} - x_{\min}}{N}$ and the time domain into M intervals of length $\Delta t = \frac{T}{M}$. Denote the points of the computational grid by $x_i = x_{\min} + i\Delta x$ for $i = 0, 1, \dots, N$, and the time steps by $t^n = n\Delta t$ for $n = 0, 1, \dots, M$. Our goal is to find an approximation $v_i^n \approx v(t^n, x_i)$ for the true solution v .

To approximate the differential operators, we rely on Taylor series expansions. Assuming v is sufficiently smooth, we have:

$$v(t, x + \Delta x) = v(t, x) + \partial_x v(t, x) \Delta x + \frac{1}{2} \partial_{xx}^2 v(t, x) \Delta x^2 + \mathcal{O}(\Delta x^3),$$

$$v(t, x - \Delta x) = v(t, x) - \partial_x v(t, x) \Delta x + \frac{1}{2} \partial_{xx}^2 v(t, x) \Delta x^2 + \mathcal{O}(\Delta x^3).$$

From these, we can derive standard finite difference operators. For the first derivative, we have the forward difference D^+ and backward difference D^- :

$$D^+ v_i^n := \frac{v_{i+1}^n - v_i^n}{\Delta x} = \partial_x v(t^n, x_i) + \mathcal{O}(\Delta x),$$

$$D^- v_i^n := \frac{v_i^n - v_{i-1}^n}{\Delta x} = \partial_x v(t^n, x_i) + \mathcal{O}(\Delta x).$$

Adding the two Taylor expansions yields the central difference approximation for the second derivative, which is second-order accurate:

$$D^2 v_i^n := \frac{v_{i+1}^n - 2v_i^n + v_{i-1}^n}{\Delta x^2} = \partial_{xx}^2 v(t^n, x_i) + \mathcal{O}(\Delta x^2).$$

When we substitute the exact, continuous solution $v(t, x)$ into our discrete finite difference operators, it will not perfectly satisfy the discrete equation. The leftover residual terms $\mathcal{O}(\Delta x)$ and $\mathcal{O}(\Delta x^2)$ from our Taylor expansions represent the error introduced strictly by moving from a continuous to a discrete domain. This residual is formalized as the local truncation error.

Definition 3.1.1 (Consistency). *Let $\mathcal{L}v = 0$ represent the continuous PDE, and let $\mathcal{L}_h v_h = 0$ represent the discretized finite difference equations with grid parameters $h = (\Delta t, \Delta x)$. The local truncation error is given by $\tau_i^n = \mathcal{L}_h v(t^n, x_i) - \mathcal{L}v(t^n, x_i)$. A numerical scheme is said to be consistent if the local truncation error vanishes as the grid spacing goes to zero:*

$$\lim_{\Delta t, \Delta x \rightarrow 0} \|\tau_i^n\| = 0.$$

Because our finite difference operators are derived directly from Taylor series expansions, this consistency property is naturally satisfied. As shown above, the first derivative approximations carry a truncation error of $\mathcal{O}(\Delta x)$ and the second derivative approximation carries an error of $\mathcal{O}(\Delta x^2)$. Assuming the true value function v is sufficiently smooth and the underlying drift and diffusion coefficients are continuous, substituting these difference operators into the discrete equation $\mathcal{L}_h v_h = 0$ reproduces the continuous operator $\mathcal{L}v$ alongside a linear combination of these $\mathcal{O}(\Delta x)$ and $\mathcal{O}(\Delta t)$ error terms. Taking the limit as the grid spacing $h = (\Delta t, \Delta x) \rightarrow 0$ forces these residual terms to vanish, confirming that our baseline discretization is mathematically consistent.

3.2 Discrete Maximum Principle and Upwind Schemes

Consistency guarantees that our discrete equations match the continuous PDE as the grid spacing shrinks to zero, but it doesn't guarantee a well-behaved solution in practice. If you apply standard central differences to a PDE dominated by strong first-derivative (or drift) terms, the numerical solution will frequently break down into large oscillations.

The numerical scheme must be physically consistent as well as mathematically consistent; it has to respect the actual flow of information. The maximum principle guarantees that, in a continuous system without internal sources, a solution will achieve its maximum or minimum strictly on the boundaries. We need the discretization to follow the same physical constraint so that we avoid these oscillations. Forcing our scheme to satisfy the Discrete Maximum Principle (DMP) naturally leads us to the algebraic concept of monotonicity.

3.2.1 Algebraic Representation and Monotone Matrices

Our discretized system may also be represented through a linear system. Suppose we discretize the spatial domain over a grid of n nodes. By replacing continuous differential operators with finite difference approximations, the PDE at each node reduces to a linear combination of the

values at adjacent nodes. Consequently, the continuous problem transforms globally into a system of linear algebraic equations.

To explicitly handle the boundary conditions, let the state vector $\mathbf{u} \in \mathbb{R}^n$ be partitioned such that the first m indices are interior nodes ($\mathbf{u}^i$) and the remaining $n - m$ indices are boundary nodes ($\mathbf{u}^b$). The full linear system can then be represented in the following block matrix form:

$$A\mathbf{u} = \begin{pmatrix} A^i & A^b \\ 0 & I \end{pmatrix} \begin{pmatrix} \mathbf{u}^i \\ \mathbf{u}^b \end{pmatrix} = \begin{pmatrix} \mathbf{g} \\ \mathbf{u}^b \end{pmatrix},$$

where the global coefficient matrix $A \in \mathbb{R}^{n \times n}$ encodes the weights of the finite difference scheme. Breaking this down, $A^i \in \mathbb{R}^{m \times m}$ corresponds to the interactions among interior nodes, $A^b \in \mathbb{R}^{m \times (n-m)}$ couples the interior nodes to the boundary nodes, I is the identity matrix enforcing the boundary values, and $\mathbf{g} \in \mathbb{R}^m$ aggregates any internal sources. The traditional approach to studying whether this system satisfies global DMPs relies on the concept of monotone matrices.

Definition 3.2.1 (Monotone Matrix). *A square matrix A is called monotone if it is non-singular and $A^{-1} \geq 0$. An equivalent characterization is that for all $\mathbf{v} \in \mathbb{R}^n$, $A\mathbf{v} \geq 0$ implies $\mathbf{v} \geq 0$.*

3.2.2 M-Matrices and the Global DMP

Computing inverse matrices is computationally expensive (most practical algorithms like Gauss-Jordan elimination are $\mathcal{O}(n^3)$), and so it is inefficient to use that to calculate if a matrix is monotone. Therefore, we restrict our scheme to a subset of monotone matrices known as M-matrices.

Definition 3.2.2 (M-Matrix). *A matrix $A = (a_{ij})_{i,j=1}^n$ is an M-matrix if it satisfies:*

1. $a_{ij} \leq 0$ for all $i \neq j$ (non-positive off-diagonal entries),
2. A is non-singular,
3. and $A^{-1} \geq 0$.

It is clear that this is a subset of the monotone matrices since conditions two and three define a monotone matrix. The third condition still requires us to invert the matrix, which doesn't solve the previous problem; however, we can prove a sufficient condition.

Definition 3.2.3 (Matrix of Non-Negative Type). *A matrix A is a matrix of non-negative type if its off-diagonal entries are non-positive ($a_{ij} \leq 0$ for $i \neq j$) and its row sums are non-negative ($\sum_{j=1}^n a_{ij} \geq 0$ for all i).*

Theorem 3.2.4. *If A is a non-singular matrix of non-negative type, then A is an M-matrix.*

Proof. By the definition of an M-matrix, we must show that A satisfies three conditions: $a_{ij} \leq 0$ for $i \neq j$, A is non-singular, and $A^{-1} \geq 0$. The first two conditions are given immediately by the definition that A is a non-singular matrix of non-negative type. Thus, we only need to show the third condition: $A^{-1} \geq 0$.

It is sufficient to show the alternative definition of the monotonicity property: for any vector $\mathbf{v}$, if $A\mathbf{v} \geq \mathbf{0}$, then $\mathbf{v} \geq \mathbf{0}$. Assume that this is not true. Then, there exists a vector $\mathbf{v}$ such that $A\mathbf{v} \geq \mathbf{0}$, but $\mathbf{v}$ has at least one strictly negative component.

Let $v_{\min} < 0$ be the smallest (most negative) value in the vector $\mathbf{v}$, and let J be the set of all indices where this minimum is achieved:

$$J = \{i : v_i = v_{\min}\}.$$

Choose any index $k \in J$. We examine the k -th component of the product $A\mathbf{v}$:

$$(A\mathbf{v})_k = a_{kk}v_k + \sum_{j \neq k} a_{kj}v_j.$$

Because A is of non-negative type, we know $a_{kj} \leq 0$ for all $j \neq k$. By our definition of J , for any $j \notin J$, we know $v_j > v_k$. Multiplying this strict inequality by the non-positive number a_{kj} yields $a_{kj}v_j \leq a_{kj}v_k$. For any $j \in J$, we have $v_j = v_k$, so $a_{kj}v_j = a_{kj}v_k$. Substituting these into

our equation yields:

$$(A\mathbf{v})_k \leq a_{kk}v_k + \sum_{j \neq k} a_{kj}v_k = v_k \left(\sum_{j=1}^n a_{kj} \right).$$

Because A is of non-negative type, its row sums are non-negative ($\sum_{j=1}^n a_{kj} \geq 0$). Because $v_k < 0$, multiplying the non-negative row sum by the strictly negative v_k guarantees a value less than or equal to zero:

$$(A\mathbf{v})_k \leq 0.$$

However, our initial assumption was that $A\mathbf{v} \geq \mathbf{0}$, meaning every component must be non-negative. Since we have $(A\mathbf{v})_k \geq 0$ and $(A\mathbf{v})_k \leq 0$, we have that:

$$(A\mathbf{v})_k = 0.$$

This requires two things to be true for row k :

1. The row sum must be exactly zero: $\sum_{j=1}^n a_{kj} = 0$ because we have that $v_k < 0$ and $0 = (A\mathbf{v})_k \leq v_k(\sum_{j=1}^n a_{kj}) \leq 0$.
2. For all $j \notin J$, it must be that $a_{kj} = 0$ because $a_{kk}v_k + \sum_{j \neq k} a_{kj}v_j = v_k \sum_{j=1}^n a_{kj}$ and $v_j > v_k$.

Because $a_{kj} = 0$ for all $k \in J$ and $j \notin J$, the variables in the set J are completely decoupled from the rest of the grid. If we permute the indices so that the elements of J appear first, the matrix A takes a block lower-triangular form:

$$A = \begin{pmatrix} A_{JJ} & \mathbf{0} \\ A_{J^c J} & A_{J^c J^c} \end{pmatrix},$$

where A_{JJ} is the square submatrix corresponding to the indices in J , and so the block to the right of A_{JJ} is entirely zeros. Thus, the row sums of the submatrix A_{JJ} are identical to the row sums of the full matrix A for those indices, which must be 0.

We then have that the vector of ones $\mathbf{1}$ is in the null space of A_{JJ} because $A_{JJ}\mathbf{1} = \mathbf{0}$, meaning the submatrix A_{JJ} is singular ($\det(A_{JJ}) = 0$). The determinant of a block triangular matrix is the product of the determinants of its diagonal blocks:

$$\det(A) = \det(A_{JJ}) \times \det(A_{JC JC}).$$

Since $\det(A_{JJ}) = 0$, it must be that $\det(A) = 0$. This implies that the entire matrix A is singular, which directly contradicts our initial premise that A is a non-singular matrix. Therefore, our assumption is false, and the vector $\mathbf{v}$ cannot have any negative components, confirming that it is monotone and that $A^{-1} \geq 0$. Because A has non-positive off-diagonals, is non-singular, and has a non-negative inverse, it is an M-matrix. $\square$

From Levy–Desplanques [15], we obtain the following theorem:

Theorem 3.2.5 (Levy-Desplanques Theorem). *Strictly diagonally dominant matrices are non-singular, which is to say that if $A \in \mathbb{C}^{n,n}$ is a matrix satisfying the property*

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|$$

for all i , then $\det(A) \neq 0$.

Proof. Assume that this is not true, meaning that a strictly diagonally dominant matrix A has $\det(A) = 0$. Then, there exists a non-zero vector $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{0}$. Define M to be the index where $\mathbf{x}$ achieves its maximum:

$$|x_M| = \max(|x_1|, \dots, |x_n|).$$

We then have that:

$$a_{M1}x_1 + \dots + a_{MM}x_M + \dots + a_{Mn}x_n = 0,$$

which implies

$$|a_{MM}||x_M| = |a_{MM}x_M| = \left| \sum_{j \neq M} a_{Mj}x_j \right| \leq \sum_{j \neq M} |a_{Mj}||x_j| \leq |x_M| \sum_{j \neq M} |a_{Mj}|,$$

but $|a_{MM}| \leq \sum_{j \neq M} |a_{Mj}|$ directly contradicts the strictly diagonally dominant assumption that $|a_{MM}| > \sum_{j \neq M} |a_{Mj}|$. Thus, we must have that $\det(A) \neq 0$. $\square$

By combining theorems 3.2.4 and 3.2.5, we can efficiently secure the monotonicity of our scheme by carefully designing the signs and magnitudes of the entries in our discretized linear system. If we generate a strictly diagonally dominant matrix of non-negative type, we get a monotone matrix, bypassing the need to compute the inverse. To guarantee that our discrete solution satisfies the global DMP, we rely on the following theorem connecting monotone matrices to the DMP.

Recall that our state vector $\mathbf{u}$ (which represents the numerical value function $\mathbf{v}^n$ at a given time step) is partitioned into interior nodes (indices 1 through m) and boundary nodes (indices $m+1$ through n). The mathematical condition $\sum_{j=1}^n a_{ij}u_j \leq 0$ corresponds to evaluating our discretized PDE operator at an interior node without any internal sources.

Theorem 3.2.6 (Monotone Matrices and the Global DMP). *If the matrix A is monotone and also satisfies the row-sum condition:*

$$\sum_{j=1}^n a_{ij} \geq 0 \quad \text{for } 1 \leq i \leq m,$$

then the discrete solution satisfies the global DMP:

$$\sum_{j=1}^n a_{ij}u_j \leq 0, \quad i = 1, \dots, m \implies \max_{i=1, \dots, n} u_i \leq \max_{i=m+1, \dots, n} u_i^+,$$

where $u_i^+ = \max\{u_i, 0\}$.

The theorem establishes that if the matrix is monotone and has non-negative row sums, the

maximum value of the entire solution vector $\mathbf{u}$ cannot exceed the maximum value found strictly on its boundaries (represented by $\max u_i^+$ for indices $m + 1$ to n). Intuitively, the Discrete Maximum Principle dictates that a quantity cannot spontaneously generate new peaks or valleys within the interior of the domain; the extreme values must be inherited from the boundary conditions or the terminal payoff. By securing the algebraic condition $\max u_i \leq \max u_{\text{boundary}}$, we guarantee that information flows logically from the boundaries inward, preserving the realism of the approximation.

3.2.3 Upwind Differencing

To ensure that our discretization yields a monotone matrix and thus satisfies the global DMP, we must design the finite difference operators to meet the sign property $a_{ij} \leq 0$ for $i \neq j$.

When approximating a drift term $b(t, x)\partial_x v$, central differences can produce positive off-diagonal entries if the drift dominates the diffusion, violating the M-matrix condition and causing oscillations. To preserve the M-matrix structure, we replace central differences with an upwind difference scheme:

$$\partial_x v \approx \begin{cases} D^+ v_i^n = \frac{v_{i+1}^n - v_i^n}{\Delta x} & \text{if } b(t_n, x_i) < 0 \\ D^- v_i^n = \frac{v_i^n - v_{i-1}^n}{\Delta x} & \text{if } b(t_n, x_i) > 0 \end{cases}.$$

By looking “upwind” into the direction of the flow of information, the coefficient corresponding to the neighboring node is always non-positive. This ensures that the off-diagonal elements of the linear system remain non-positive. In the next section, we will show that the time-stepping scheme completes the requirements to form an M-matrix and satisfy the discrete maximum principle.

3.3 Time-Stepping and Stability

The HJB equation is a terminal value problem since we are given the final payoff $g(x)$ at $t = T$, and so we must step backwards in time from T to 0. We have two primary choices for handling

the time derivative $\partial_t v$: explicit methods and implicit methods.

An explicit scheme evaluates the spatial derivatives at a known future time t^{n+1} to solve for the current time t^n :

$$\frac{v_i^{n+1} - v_i^n}{\Delta t} + \mathcal{L}_h v_i^{n+1} = 0 \implies v_i^n = v_i^{n+1} + \Delta t (\mathcal{L}_h v_i^{n+1}).$$

Explicit schemes are typically much less computationally expensive as there are no complicated matrix inversions or decompositions, but they are subject to harsh stability restrictions.

We could also consider the backward Euler scheme, which is an implicit method. In an implicit scheme, the spatial derivatives are evaluated at the unknown current time t^n :

$$\frac{v_i^{n+1} - v_i^n}{\Delta t} + \mathcal{L}_h v_i^n = 0 \implies \mathbf{v}^n = (\mathcal{I} - \Delta t \mathcal{L}_h)^{-1} \mathbf{v}^{n+1},$$

where $\mathcal{I}$ is the identity matrix and $\mathcal{L}_h$ is the spatial difference operator matrix. This scheme does require matrix inversion, which means it is much slower; however, it is subject to less stability requirements.

Definition 3.3.1 (Stability). *A finite difference scheme is stable if the numerical solution remains uniformly bounded as $\Delta t, \Delta x \rightarrow 0$ for a fixed terminal time T .*

For explicit schemes applied to parabolic PDEs, Von Neumann stability analysis yields the Courant-Friedrichs-Lewy condition [6].

Theorem 3.3.2 (Courant-Friedrichs-Lewy Condition). *Let us be given a general advection-diffusion equation with a drift coefficient $b(t, x)$ and a diffusion coefficient $\frac{1}{2}\sigma^2(t, x)$. To solve this using a finite difference approach, the time step Δt must be strictly bounded by the spatial step Δx to ensure non-negative transition probabilities. Specifically, the grid parameters must satisfy the inequality:*

$$\Delta t \leq \left(\frac{|b(t, x)|}{\Delta x} + \frac{\sigma^2(t, x)}{\Delta x^2} \right)^{-1}$$

everywhere in the computational domain.

This forces the time step to be impractically small to satisfy stability in a multi-dimensional setting with fine grids. Thus, we turn away from explicit methods and look to implicit methods to stabilize our scheme. In choosing an implicit method, the global matrix takes the form $A = \mathcal{I} - \Delta t \mathcal{L}_h$. The identity matrix adds 1 to every diagonal element. Because the off-diagonal elements of $-\Delta t \mathcal{L}_h$ are non-positive and their absolute sum is scaled by Δt , the diagonal element remains strictly greater than the sum of the off-diagonals, securing both strict diagonal dominance and non-negative row sums, completing the DMP requirements. From this scheme, we also obtain:

Theorem 3.3.3. *The fully implicit finite difference scheme for parabolic PDEs is unconditionally stable in the L^∞ -norm, regardless of the choice of Δt and Δx .*

Proof. This is a direct consequence of the DMP. Morton and Mayers [12, Theorem 2.2, p. 34] show that the θ -method satisfies the maximum principle and is stable in the maximum norm whenever the grid ratio satisfies $\frac{\Delta t}{\Delta x^2}(1 - \theta) \leq \frac{1}{2}$. For the backward Euler scheme, $\theta = 1$, and so this is unconditionally satisfied for all $\Delta t, \Delta x > 0$. $\square$

3.4 Convergence and the Barles-Souganidis Theorem

Classical solutions to HJB equations often do not exist because they assume the true solution v is in $\mathcal{C}^{1,2}$. When the true solution lacks this regularity, we must rely on a more generalized framework to guarantee that our numerical method actually converges to the correct value function. For numerical analysis on fully nonlinear second-order PDEs, the Barles-Souganidis theorem [3] utilizes the concept of viscosity solutions to work around this issue. The Weak Dynamic Programming Principle (2.3.4) defines the upper and lower semicontinuous envelopes of a locally bounded function v :

$$v^*(t, x) := \limsup_{(t', x') \rightarrow (t, x)} v(t', x'), \quad \text{and} \quad v_*(t, x) := \liminf_{(t', x') \rightarrow (t, x)} v(t', x').$$

We can use smooth envelopes $\psi \in \mathcal{C}^{1,2}([0, T] \times \mathbb{R})$ to approach the true value function v to avoid discontinuities.

Definition 3.4.1 (Viscosity Solution of the HJB Equation). *A locally bounded function v is a viscosity subsolution to the HJB equation if, for all test functions $\psi \in \mathcal{C}^{1,2}([0, T] \times \mathbb{R})$ and points (t, x) where $v^* - \psi$ achieves a local maximum, we have:*

$$-\partial_t \psi(t, x) - \mathcal{H}(t, x, v^*(t, x), \partial_x \psi(t, x), \partial_{xx}^2 \psi(t, x)) \leq 0.$$

Conversely, v is a viscosity supersolution if, for all test functions $\psi \in \mathcal{C}^{1,2}([0, T] \times \mathbb{R})$ and points (t, x) where $v_ - \psi$ achieves a local minimum, we have:*

$$-\partial_t \psi(t, x) - \mathcal{H}(t, x, v_*(t, x), \partial_x \psi(t, x), \partial_{xx}^2 \psi(t, x)) \geq 0.$$

A function v is a viscosity solution of the HJB equation if it is both a viscosity subsolution and a viscosity supersolution.

We introduce a discretization scheme for a numerical solution. Let $h = (\Delta t, \Delta x)$ represent the grid spacing parameters. We can write the finite difference scheme abstractly as an operator S :

$$S(h, t, x, v^h(t, x), v^h) = 0,$$

where v^h is the numerical approximation generated by the scheme. The Barles-Souganidis Theorem (1991) tells us that the operator S must satisfy three fundamental conditions to guarantee convergence:

1. **Monotonicity:** The discrete scheme must preserve the ordering of solutions. For any grid parameter h , point (t, x) , constant $r \in \mathbb{R}$, and bounded functions u, w such that $u \geq w$ everywhere, the operator satisfies:

$$S(h, t, x, r, u) \leq S(h, t, x, r, w).$$

2. **Stability:** For all h , the numerical scheme admits a solution v^h that is uniformly bounded independently of h .
3. **Consistency:** As the grid spacing shrinks, the discrete operator must recover the continuous HJB operator for any smooth test function $\psi \in \mathcal{C}^{1,2}([0, T] \times \mathbb{R})$. Specifically, as $h \rightarrow 0$, $(s, y) \rightarrow (t, x)$, and $\epsilon \rightarrow 0$, the limits of $\frac{1}{h}S(h, s, y, \psi(s, y) + \epsilon, \psi + \epsilon)$ are bounded below and above by the continuous HJB operator evaluated at $\psi(t, x)$.

Theorem 3.4.2 (Barles-Souganidis Convergence Theorem). *Assume the numerical scheme S is monotone, stable, and consistent. Furthermore, assume the continuous HJB equation satisfies a strong comparison principle: if u is an upper semicontinuous subsolution and w is a lower semicontinuous supersolution, then $u \leq w$ everywhere.*

Under these conditions, as the grid parameters $h \rightarrow 0$, the numerical solutions v^h converge locally uniformly to the unique continuous viscosity solution v of the HJB equation.

Proof. Let the upper and lower semicontinuous envelopes for the numerical solution be defined as:

$$\begin{aligned}\bar{v}(t, x) &:= \limsup_{h \rightarrow 0, (s, y) \rightarrow (t, x)} v^h(s, y), \\ \underline{v}(t, x) &:= \liminf_{h \rightarrow 0, (s, y) \rightarrow (t, x)} v^h(s, y).\end{aligned}$$

We claim that $\bar{v}$ is a viscosity subsolution and $\underline{v}$ is a viscosity supersolution to the HJB equation. If this claim holds, the strong comparison principle tells us that $\bar{v} \leq \underline{v}$ everywhere. By their definitions as supremum and infimum limits, the opposite inequality $\underline{v} \leq \bar{v}$ is also true. Therefore, $\bar{v} = \underline{v} = v$ is the unique continuous viscosity solution, and this equality guarantees that the sequence v^h converges locally uniformly to v .

First, let's prove the claim for the subsolution $\bar{v}$. Let $\psi \in \mathcal{C}^{1,2}([0, T] \times \mathbb{R})$ be an arbitrary smooth test function, and suppose (t_0, x_0) is a strict local maximum of $\bar{v} - \psi$. Without loss of generality, we may assume $\bar{v}(t_0, x_0) = \psi(t_0, x_0)$ (otherwise they have a constant gap, and so the general ψ and this new ψ without the gap have the same partial derivatives). We can also modify ψ outside

a small neighborhood $B((t_0, x_0), r)$ such that $\psi \geq 2 \sup_h \|v^h\|_\infty$ because the maximum is strict. This guarantees that $\bar{v} - \psi < 0$ outside the neighborhood. Thus, there exist sequences of grid parameters $h_n \rightarrow 0$ and points (t_n, x_n) such that as $n \rightarrow \infty$, $v^{h_n}(t_n, x_n) \rightarrow \bar{v}(t_0, x_0)$. Furthermore, because ψ was made arbitrarily large outside the neighborhood, the global maximum of $v^{h_n} - \psi$ is forced to occur inside the neighborhood, meaning (t_n, x_n) is a global maximum point and $(t_n, x_n) \rightarrow (t_0, x_0)$.

Let $\xi_n = v^{h_n}(t_n, x_n) - \psi(t_n, x_n)$. Observe that $\xi_n \rightarrow 0$ because $v^{h_n}(t_n, x_n)$ converges to $\bar{v}(t_0, x_0)$ and ψ is continuous. Furthermore, because (t_n, x_n) is a global maximum, we have

$$v^{h_n}(s, y) \leq \psi(s, y) + \xi_n$$

for all (s, y) . We have that $S(h_n, t_n, x_n, v^{h_n}(t_n, x_n), v^{h_n}) = 0$, and so we can apply the scheme's monotonicity property, which preserves the previous inequality:

$$S(h_n, t_n, x_n, \psi(t_n, x_n) + \xi_n, \psi + \xi_n) \leq 0.$$

Dividing by h_n and taking the limit inferior as $n \rightarrow \infty$, we invoke the consistency condition of the scheme:

$$\begin{aligned} 0 &\geq \liminf_{n \rightarrow \infty} \frac{S(h_n, t_n, x_n, \psi(t_n, x_n) + \xi_n, \psi + \xi_n)}{h_n}, \\ 0 &\geq -\partial_t \psi(t_0, x_0) - \mathcal{H}(t_0, x_0, \psi(t_0, x_0), \partial_x \psi(t_0, x_0), \partial_{xx}^2 \psi(t_0, x_0)). \end{aligned}$$

Because we assumed $\bar{v}(t_0, x_0) = \psi(t_0, x_0)$, this gives:

$$-\partial_t \psi(t_0, x_0) - \mathcal{H}(t_0, x_0, \bar{v}(t_0, x_0), \partial_x \psi(t_0, x_0), \partial_{xx}^2 \psi(t_0, x_0)) \leq 0.$$

Thus, $\bar{v}$ is a subsolution. The proof for the supersolution $\underline{v}$ follows a symmetric argument, yielding the desired result. $\square$

By this theorem, the previous numerical scheme choices connect back to stochastic optimal

control. We secure consistency, monotonicity, and stability through the choice of finite difference operators, an upwind scheme, and an implicit time-stepper. This fulfills the theoretical conditions of the Barles-Souganidis theorem, guaranteeing that the output of our numerical solver converges to the unique viscosity solution for an HJB meeting sufficient conditions.

3.5 Handling Nonlinearities via Linearization

A final computational challenge arises from the inherent nonlinearity of the HJB equation since it features a supremum operator over the control and often quadratic terms. A fully implicit treatment of a nonlinear term like $(v_x)^2$ or a^2 requires solving a massive system of nonlinear algebraic equations at every time step using algorithms like Newton-Raphson, which is computationally expensive. Instead, we consider a semi-implicit Picard iteration.

Suppose the PDE contains a nonlinear term $F(v_i^n)$. We approximate this by lagging one part of the nonlinearity from a previous iteration k :

$$F(v_i^{n,(k+1)}) \approx \tilde{F}(v_i^{n,(k)}) \cdot v_i^{n,(k+1)}$$

Under the assumptions of the Banach fixed-point theorem [2], this maintains the convergence of our sequence.

Theorem 3.5.1 (Banach Fixed-Point Theorem). *Let X be a complete metric space and $T : X \rightarrow X$ be a contraction mapping, meaning there exists a constant $L \in [0, 1)$ such that $d(T(x), T(y)) \leq L \cdot d(x, y)$ for all $x, y \in X$. Then T admits a unique fixed point $x^* \in X$, and the sequence $x^{(k+1)} = T(x^{(k)})$ converges to x^* .*

By framing our semi-implicit time step as a mapping $T(\mathbf{v}^{(k)}) = \mathbf{v}^{(k+1)}$, we can linearly decouple the PDE. As long as the time step Δt is sufficiently small to ensure T remains a contraction mapping, iterating the linear system will converge to the true solution of the fully nonlinear step, combining the stability of implicit methods with the speed of linear sparse matrix solvers.

Chapter 4

Optimal Trading in Commodities

Now that we have reviewed the relevant background material, we can introduce the problem as proposed in Chan et al. [5]. The authors formulate a dynamic portfolio optimization model for equities that accounts for market frictions alongside predictable returns and multiscale stochastic volatility. The problem seeks to determine an optimal, continuous trading trajectory over a given horizon, balancing the drive to maximize expected returns against risk aversion and the friction costs incurred by trading. Rather than a static buy-and-hold allocation, the investor dynamically adjusts their position based on the optimal rate of trading at any given instant. Here, we extend this framework to commodity markets. By mapping the problem to commodity price dynamics via the Schwartz two-factor model, we adapt their approach to capture the unique structures of commodity markets.

4.1 Market Trading Model

For simplicity, we will assume that traders can trade the commodity and its spot price directly. In reality, directly trading the spot commodity is not practical for financial market participants due to the substantial costs associated with storage, transportation, and physical delivery. The spot price effectively serves as a theoretical variable. Instead, market participants primarily gain exposure to these markets through derivative instruments, most notably futures contracts.

4.1.1 Frictions

I adopt the same model of trading frictions as in Chan et al. Namely, let q_t denote the size of the position and let u_t denote the trading rate at time t so that:

$$dq_t = u_t dt.$$

The transaction costs may then be modeled as:

$$TC = \frac{K}{2} u_t^2,$$

where K is a constant > 0 . Transaction costs are assumed to be instantaneous (i.e. liquidity will revert back to original level). The original paper also includes price impact, which is assumed to be persistent. If we model the price as

$$S_t = P_t + l_t,$$

then S_t is the perturbed price (after price impact), P_t is the unperturbed price (spot price of the commodity), and l_t is the price impact. However, we will drop this assumption for simplicity.

4.1.2 Price Dynamics

I assume the standard two-factor model as developed by Schwartz [14]. It posits that the commodity spot price is driven by two unobservable, stochastic factors: a long-term equilibrium price level and a short-term deviation (or convenience yield). More specifically, it takes

$$X_t = \ln(P_t),$$

so that X_t is the log spot price. From there, we get:

$$dX_t = \left(\mu - \delta_t - \frac{\sigma_X(Y_t, Z_t)^2}{2} \right) dt + \sigma_X(Y_t, Z_t) dW_t^{(1)},$$

where $W_t^{(1)}$ is a standard Brownian motion, μ is a constant representing the drift, δ_t is the convenience yield, and $\sigma_X(Y_t, Z_t)$ is the volatility. The convenience yield δ_t has the dynamics:

$$d\delta_t = \kappa(\theta - \delta_t) dt + \sigma_\delta dW_t^{(2)},$$

where $W_t^{(2)}$ is a Brownian motion with $\rho_{12} = \text{corr}(dW_t^{(1)}, dW_t^{(2)})$, θ is a constant representing the long run mean, κ is a constant representing the speed of mean reversion, and σ_δ is the volatility. We assume a multiscale stochastic volatility framework in which the spot-price volatility depends on a fast factor Y_t and a slow factor Z_t , so that $\sigma_X = \sigma_X(Y_t, Z_t)$. These factors evolve as

$$\begin{aligned} dY_t &= \frac{1}{\epsilon} b(Y_t) dt + \frac{1}{\sqrt{\epsilon}} a(Y_t) dW_t^{(3)}, \\ dZ_t &= \delta c(Z_t) dt + \sqrt{\delta} g(Z_t) dW_t^{(4)}, \end{aligned}$$

where ϵ and δ govern the fast and slow time scales, respectively. The Brownian motions $W_t^{(1)}, W_t^{(2)}, W_t^{(3)}, W_t^{(4)}$ drive the spot price, convenience yield, fast volatility factor, and slow volatility factor, respectively. We allow $W_t^{(1)}$ and $W_t^{(2)}$ to be correlated, and also allow correlation among the volatility-related drivers $W_t^{(2)}, W_t^{(3)}, W_t^{(4)}$.

By Itô's formula, we may write the original price dynamics:

$$dP_t = P_t(\mu - \delta_t) dt + P_t \sigma_X(Y_t, Z_t) dW_t^{(1)}.$$

Finally, let's take $x_t = \mu - \delta_t$ to be our return-predictive signal so that:

$$dP_t = P_t x_t dt + P_t \sigma_X(Y_t, Z_t) dW_t^{(1)},$$

$$dx_t = \kappa(\mu - \theta - x_t) dt - \sigma_\delta dW_t^{(2)}.$$

4.1.3 Objective

The Schwartz two-factor model switches the price dynamics to a geometric diffusion, which introduces nonlinearity into the HJB that wasn't present in Chan et al.'s original paper. Because of this nonlinearity, we seek a numerical solution. We will also switch from the infinite horizon problem of the original paper to a terminal horizon problem for purposes of the time-stepping algorithm.

Our wealth function incorporates the trader's position in the commodity, avoidance of risk exposure, and transaction costs. Profit and loss from the strategy follow:

$$\Pi_s = \int_0^s q_t dS_t = \int_0^s q_t dP_t.$$

Substituting our asset price dynamics $dP_t = P_t x_t dt + P_t \sigma_X(Y_t, Z_t) dW_t^{(1)}$, the expected incremental wealth is simply the drift component, as the Brownian increment vanishes under the expectation operator:

$$\mathbb{E}[d\Pi_s] = \mathbb{E}[q_s P_s x_s ds].$$

This term, $q_s P_s x_s$, represents the expected excess return (or “alpha”) generated by the predictable signal x_s . The trader is not risk-neutral, though. For a mean-variance objective, we introduce a continuous penalty for the variance of the position. The instantaneous variance of the position value is $(q_s P_s \sigma_X(Y_s, Z_s))^2$. Applying a risk-aversion parameter γ , the continuous risk penalty becomes:

$$-\frac{\gamma}{2} q_s^2 P_s^2 \sigma_X(Y_s, Z_s)^2 ds.$$

Additionally, we bring back the instantaneous transaction costs $-\frac{K}{2} u_s^2 ds$ and include a liquidation penalty $g(q_T, P_T, x_T, Y_T, Z_T)$, which represents the cost of unwinding any residual inventory q_T at the terminal date T . Similarly to the transaction costs, we will assume a quadratic liqui-

dition penalty $g(q_T, P_T, x_T, Y_T, Z_T) = -\frac{\alpha_T}{2}q_T^2$, where α_T is a constant. Finally, we discount the cash flows at a continuous rate $\rho > 0$ to obtain the trader's value function:

$$v(t, q, P, x, y, z) = \sup_{\{u_s\}_{s \in [t, T]}} \mathbb{E}_{t, q, P, x, y, z} \left[\int_t^T e^{-\rho(s-t)} \left(q_s P_s x_s - \frac{\gamma}{2} q_s^2 P_s^2 \sigma_X(Y_s, Z_s)^2 - \frac{K}{2} u_s^2 \right) ds \right. \\ \left. + e^{-\rho(T-t)} g(q_T, P_T, x_T, Y_T, Z_T) \right].$$

4.1.4 Hamilton-Jacobi-Bellman Equation

We obtain the following HJB equation:

$$-v_t = -\rho v + \sup_u \left\{ -\frac{K}{2} u^2 + uv_q \right\} + qPx - \frac{\gamma}{2} q^2 P^2 \sigma_X(y, z)^2 + Pxv_P + \frac{1}{2} \sigma_X(y, z)^2 P^2 v_{PP} \\ + \kappa(\mu - \theta - x)v_x + \frac{1}{2} \sigma_\delta^2 v_{xx} - \rho_{12} \sigma_X(y, z) \sigma_\delta P v_{Px} + \left(\frac{1}{\epsilon} L_0 v + \frac{1}{\sqrt{\epsilon}} L_1 v \right) + \sqrt{\delta} M_1 v + \delta M_2 v + \sqrt{\frac{\delta}{\epsilon}} M_3 v,$$

where the infinitesimal generators and mixed-derivative operators are

$$L_0 = \frac{1}{2} a(y)^2 \frac{\partial^2}{\partial y^2} + b(y) \frac{\partial}{\partial y},$$

$$L_1 = -\sigma_\delta \rho_{23} a(y) \frac{\partial^2}{\partial x \partial y},$$

$$M_1 = -\sigma_\delta \rho_{24} g(z) \frac{\partial^2}{\partial x \partial z},$$

$$M_2 = \frac{1}{2} g(z)^2 \frac{\partial^2}{\partial z^2} + c(z) \frac{\partial}{\partial z},$$

$$M_3 = \rho_{34} a(y) g(z) \frac{\partial^2}{\partial y \partial z}.$$

The maximization problem is quadratic in u , which yields the optimal trading rate:

$$u^* = \frac{v_q}{K}.$$

Substituting this back into the HJB simplifies the supremum term to $\frac{1}{2K}v_q^2$. For notational convenience, we group the spatial derivatives into the operator L_{SV} :

$$\begin{aligned} L_{SV}f := & Px f_P + \frac{1}{2}\sigma_X(y, z)^2 P^2 f_{PP} + \kappa(\mu - \theta - x)f_x + \frac{1}{2}\sigma_\delta^2 f_{xx} - \rho_{12}\sigma_X(y, z)\sigma_\delta P f_{Px} \\ & + \left(\frac{1}{\epsilon}L_0 f + \frac{1}{\sqrt{\epsilon}}L_1 f \right) + \sqrt{\delta}M_1 f + \delta M_2 f + \sqrt{\frac{\delta}{\epsilon}}M_3 f, \end{aligned}$$

giving the condensed PDE:

$$-v_t = -\rho v + \frac{1}{2K}v_q^2 + qPx - \frac{\gamma}{2}q^2 P^2 \sigma_X(y, z)^2 + L_{SV}v$$

with terminal condition $v(T, \cdot) = -\frac{\alpha_T}{2}q^2$.

4.1.5 Ansatz

To solve this system, we use the following quadratic ansatz in the inventory variable q :

$$v(t, q, P, x, y, z) = a(t, P, x, y, z)q^2 + b(t, P, x, y, z)q + c(t, P, x, y, z).$$

By substituting this ansatz into the HJB equation and matching the coefficients for powers of q , the problem decomposes into a system of three coupled partial differential equations:

$$\begin{aligned} -a_t &= -\rho a + \frac{2}{K}a^2 - \frac{\gamma}{2}\sigma_X(y, z)^2 P^2 + L_{SV}a, \\ -b_t &= -\rho b + \frac{2}{K}ab + Px + L_{SV}b, \\ -c_t &= -\rho c + \frac{1}{2K}b^2 + L_{SV}c, \end{aligned}$$

with the terminal conditions:

$$a(T, P, x, y, z) = -\frac{\alpha_T}{2}, \quad b(T, P, x, y, z) = 0, \quad c(T, P, x, y, z) = 0.$$

4.2 Numerical Analysis

First, let us check the strong comparison property needed for the Barles–Souganidis convergence theorem. After the quadratic ansatz

$$v(t, q, P, x, y) = a(t, P, x, y)q^2 + b(t, P, x, y)q + c(t, P, x, y),$$

the numerical scheme is applied to the coefficient equations for a , b , and c on a truncated bounded domain in (P, x, y) . We drop the second volatility factor because of data availability, and we take $\sigma_X(y) = \sqrt{y}$ so that y represents the instantaneous variance. Under this reduction, the a -equation may be written in the form

$$-a_t + F_a(\xi, a, Da, D^2a) = 0, \quad \xi = (P, x, y),$$

where

$$F_a(\xi, s, p, X) = -\text{Tr}(A(\xi)X) - \beta(\xi) \cdot p + \rho s - \frac{2}{K}s^2 + \frac{\gamma}{2}yP^2.$$

Here $A(\xi)$ is the diffusion matrix and $\beta(\xi)$ collects the drift terms. Since the coefficients are continuous on the truncated bounded domain, F_a is continuous. Also, because $A(\xi)$ is positive semidefinite, the operator is degenerate elliptic: if $X \leq Y$ in the matrix order, then

$$F_a(\xi, s, p, Y) \leq F_a(\xi, s, p, X).$$

The only subtle point is the reaction term

$$s \mapsto \rho s - \frac{2}{K}s^2.$$

For the comparison principle, we need the operator to be proper, meaning nondecreasing in the scalar argument s . Globally this is not true for all $s \in \mathbb{R}$, since the quadratic term is concave.

However, the relevant solution region is $s \leq 0$. Indeed, the terminal condition satisfies

$$a(T, \cdot) = -\frac{\alpha_T}{2} \leq 0,$$

and the constant function 0 is a supersolution, since substituting $a \equiv 0$ into the PDE leaves the nonnegative term

$$\frac{\gamma}{2}yP^2 \geq 0.$$

Thus, by the parabolic maximum principle, any solution satisfies $a \leq 0$ throughout the domain.

On this region, the reaction term is increasing because

$$\frac{d}{ds} \left(\rho s - \frac{2}{K}s^2 \right) = \rho - \frac{4}{K}s \geq \rho > 0 \quad \text{for all } s \leq 0.$$

So although the nonlinearity is not globally monotone, it is monotone on the invariant region where the solution actually lives. Therefore the operator for the a -equation is proper, continuous, and degenerate elliptic on the bounded computational domain, which is exactly the structure needed for the strong comparison principle. Once a is fixed and bounded, the equations for b and c become linear parabolic equations with continuous coefficients and positive zeroth-order terms, so comparison holds for them as well. Hence the reduced coefficient system satisfies the strong comparison principle on the bounded computational domain.

Since the finite-difference scheme from the previous section is monotone, stable, and consistent, the Barles–Souganidis theorem implies convergence of the numerical approximations to the unique viscosity solution of this reduced PDE system. The numerical analysis is then carried out at the viscosity-solution level rather than the classical one. If the limiting solution is in fact sufficiently smooth and the feedback control recovered from it is admissible, then the verification theorem identifies that solution with the value function and the associated feedback rule with the optimal trading strategy.

Thus, I seek to discretize the PDE system for the coefficient functions a , b , and c by finite

differences, use backward Euler time-stepping to solve the equations backward from the terminal condition, and use upwind differencing for the first-order drift terms to preserve monotonicity. By the previous chapter, these choices are made precisely so that the scheme is monotone, stable, and consistent. Thus, with the comparison principle discussed above, the numerical method satisfies the hypotheses of the Barles-Souganidis theorem.

4.2.1 Data and Calibration

I utilize LLMs, specifically OpenAI's ChatGPT, for help producing the code for the numerical solver. All market data is from Bloomberg [4]. I fix our model to WTI crude oil futures data, where the nearest maturity contract acts as a proxy for the spot price. Although the theoretical model was introduced with both a fast and a slow volatility factor, the empirical implementation used here is reduced to a one-factor stochastic volatility model. In other words, the solver is implemented on the state space (P, x, y) rather than the full (P, x, y, z) space. This is a practical simplification to avoid expensive or challenging alternatives like high-frequency volatility surface data. The data used in the calibration provide a single market-based measure of volatility through OVX.

Under this reduced specification, the state variables evolve according to

$$dP_t = P_t x_t dt + \sqrt{y_t} P_t dW_t^{(P)},$$

$$dx_t = \kappa(\mu - \theta - x_t) dt + \sigma_\delta dW_t^{(x)},$$

$$dy_t = \kappa_y(\theta_y - y_t) dt + \sigma_y \sqrt{y_t} dW_t^{(y)},$$

with correlation

$$\rho_{Px} = \text{corr}(dW_t^{(P)}, dW_t^{(x)}).$$

The value function still takes the quadratic form

$$v(t, q, P, x, y) = a(t, P, x, y)q^2 + b(t, P, x, y)q + c(t, P, x, y),$$

so the optimal control becomes

$$u_t^* = \frac{v_q}{K} = \frac{2a(t, P_t, x_t, y_t)q_t + b(t, P_t, x_t, y_t)}{K}.$$

The model parameters are calibrated from the oil dataset using the front-month futures price, the six-month futures price, and the OVX implied volatility index from 1/2/19 to 4/9/26. The convenience yield is inferred from the Schwartz relation

$$\delta_t = r_t - \frac{1}{\tau} \ln \left(\frac{F_{t,T}}{P_t} \right),$$

where $\tau = 6/12$ years and r_t is proxied by the SOFR rate. Defining the predictive signal by

$$x_t = \mu - \delta_t,$$

I obtain a daily time series $\{x_n\}$, where one time step corresponds to $\Delta = 1/252$ years. I model this discrete-time signal as an AR(1) model:

$$x_{n+1} = c_x + \phi_x x_n + \varepsilon_{n+1}^x.$$

Since the continuous-time signal is modeled as the Ornstein–Uhlenbeck process

$$dx_t = \kappa(\mu - \theta - x_t) dt + \sigma_\delta dW_t^{(x)},$$

its long-run mean is $\mu - \theta$. Using an Euler discretization over one daily time step $\Delta = 1/252$,

$$x_{n+1} - x_n \approx \kappa(\mu - \theta - x_n)\Delta + \sigma_\delta \sqrt{\Delta} \eta_{n+1}^x,$$

which is equivalent to the AR(1) regression

$$x_{n+1} \approx a_x + b_x x_n + \varepsilon_{n+1}^x.$$

From the fitted coefficients, I recover

$$\kappa \approx \frac{1 - b_x}{\Delta}, \quad \theta \approx \mu - \frac{a_x}{1 - b_x},$$

and estimate the diffusion coefficient by

$$\sigma_\delta \approx \frac{\text{std}(\varepsilon^x)}{\sqrt{\Delta}}.$$

I estimate ρ_{Px} from the sample correlation between daily price innovations and AR(1) residuals of the predictive signal:

$$\rho_{Px} \approx \text{Corr}\left(\log \frac{P_{n+1}}{P_n}, \varepsilon_{n+1}^x\right).$$

To estimate the volatility factor, I use OVX as a proxy for the market's implied volatility. Writing implied volatility as

$$\sigma_t = \frac{\text{OVX}_t}{100}.$$

I define the variance factor by

$$y_t = \sigma_t^2.$$

I then model $\{y_n\}$ as a discretely observed square-root process,

$$dy_t = \kappa_y(\theta_y - y_t) dt + \sigma_y \sqrt{y_t} dW_t^{(y)}.$$

Using an Euler discretization over one daily time step $\Delta = 1/252$,

$$y_{n+1} - y_n \approx \kappa_y(\theta_y - y_n)\Delta + \sigma_y \sqrt{y_n \Delta} \eta_{n+1},$$

where η_{n+1} is approximately standard normal. This implies the discrete-time regression

$$\Delta y_n := y_{n+1} - y_n \approx \alpha_y + \beta_y y_n + \varepsilon_{n+1}^y,$$

with

$$\alpha_y = \kappa_y \theta_y \Delta, \quad \beta_y = -\kappa_y \Delta.$$

Accordingly, I estimate κ_y and θ_y from a weighted least squares regression of Δy_n on a constant and y_n , using weights proportional to $1/y_n$ to account for the heteroskedasticity implied by the square-root diffusion. The continuous-time parameters are then recovered as

$$\kappa_y \approx -\frac{\beta_y}{\Delta}, \quad \theta_y \approx -\frac{\alpha_y}{\beta_y}.$$

Finally, the volatility-of-volatility parameter σ_y is estimated from the normalized residuals:

$$\sigma_y \approx \text{std} \left(\frac{\Delta y_n - \kappa_y (\theta_y - y_n) \Delta}{\sqrt{y_n \Delta}} \right).$$

4.2.2 Algorithm Design

To solve the PDE system for a , b , and c , I use an implicit finite-difference scheme with Picard iteration for the nonlinear term on a tensor grid in (P, x, y) . The solver uses a one-year horizon

$$T = 1,$$

with daily time steps

$$\Delta t = \frac{1}{252}.$$

The spatial domain is discretized on a coarse grid with

$$N_P = 12, \quad N_x = 12, \quad N_y = 8.$$

The price grid is chosen using the empirical oil price range. The signal grid is centered around its estimated long-run mean and widened using several stationary standard deviations of the Ornstein-Uhlenbeck process. The variance grid is truncated to a compact interval away from zero in order to avoid instability near the degenerate boundary.

The fully implicit scheme is used mainly for stability. Since the equations are solved backward in time and include both drift and diffusion terms, an explicit scheme would impose a much more restrictive time-step condition. For the first-order drift terms, I use upwind differences, with the direction chosen from the sign of the local drift coefficient. For the second-order terms, I use centered differences. The mixed derivative term is also approximated by a centered stencil, and zero-normal-derivative boundary conditions are imposed at the edges of the grid.

After discretization, each backward time step produces a sparse algebraic system. The equation for a remains nonlinear because of the term

$$\frac{2}{K}a^2,$$

so I solve it iteratively by freezing the nonlinear term and repeating the sparse solve until convergence. Once a is updated, the equations for b and c become linear and can be solved sequentially. In the implementation, I also enforce that a remains non-positive, since this preserves concavity of the value function in the inventory variable and keeps the feedback control well behaved.

Once the coefficient functions have been computed, I evaluate the resulting strategy by Monte Carlo simulation. The state variables (P_t, x_t, y_t) are simulated forward with an Euler-Maruyama scheme, while the inventory evolves according to

$$q_{t+\Delta t} = q_t + u_t^* \Delta t,$$

where

$$u_t^* = \frac{2a(t, P_t, x_t, y_t)q_t + b(t, P_t, x_t, y_t)}{K}.$$

The trader's cash account is updated along each path by subtracting execution costs and temporary transaction costs, and terminal PnL is computed after applying the liquidation penalty. To evaluate the effect of stochastic volatility, I compare this strategy to a baseline constant-volatility strategy in which the variance factor is frozen at its long-run mean. For the main experiments, I simulate 5000 paths for each strategy, and I repeat the comparison across 2000 paths for several values of the risk-aversion parameter γ .

4.2.3 Numerical Results

With the calibration, we can examine how the model performs on the Monte Carlo simulations.

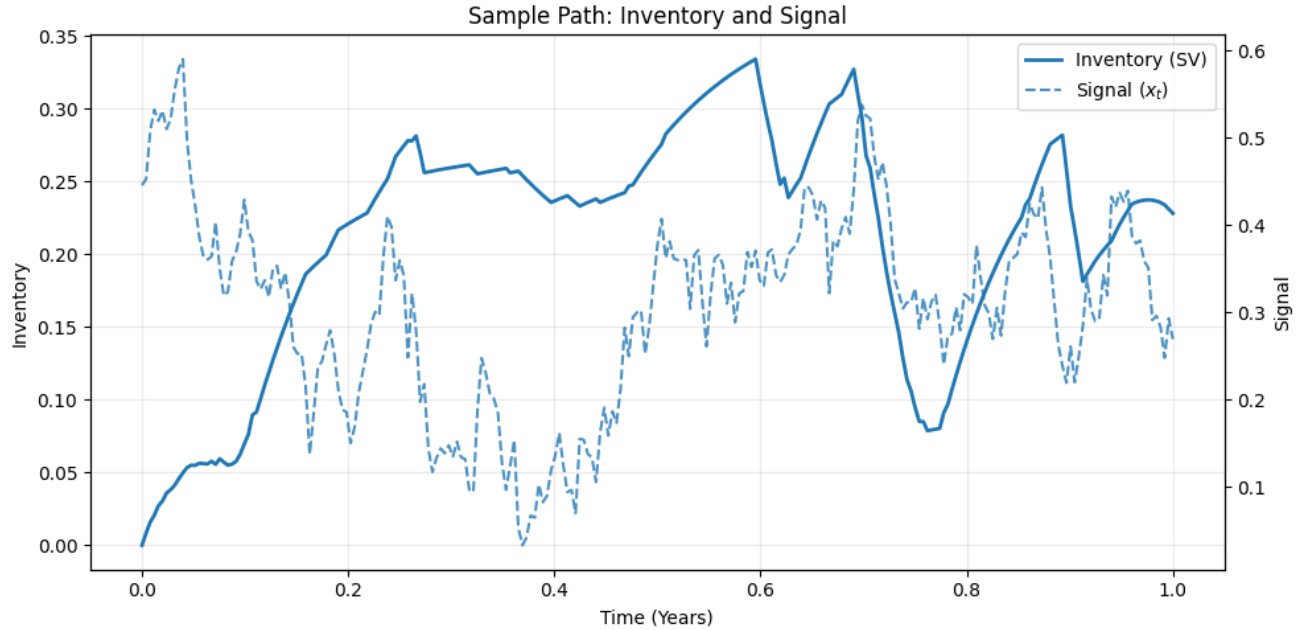


Figure 4.1: Sample path of optimal inventory and predictive signal under the stochastic-volatility strategy.

Figure 4.1 shows a sample path of the inventory together with the predictive signal. The signal fluctuates considerably over time, and the inventory adjusts accordingly but less drastically. This is what we would expect from the control problem. Even when the signal becomes more favorable, the trader does not immediately move to a large position, since doing so would incur higher transaction costs and greater risk exposure. Instead, the position is adjusted gradually.

Thus, the figure illustrates how the optimal strategy smooths the raw signal through the presence of frictions and risk aversion.

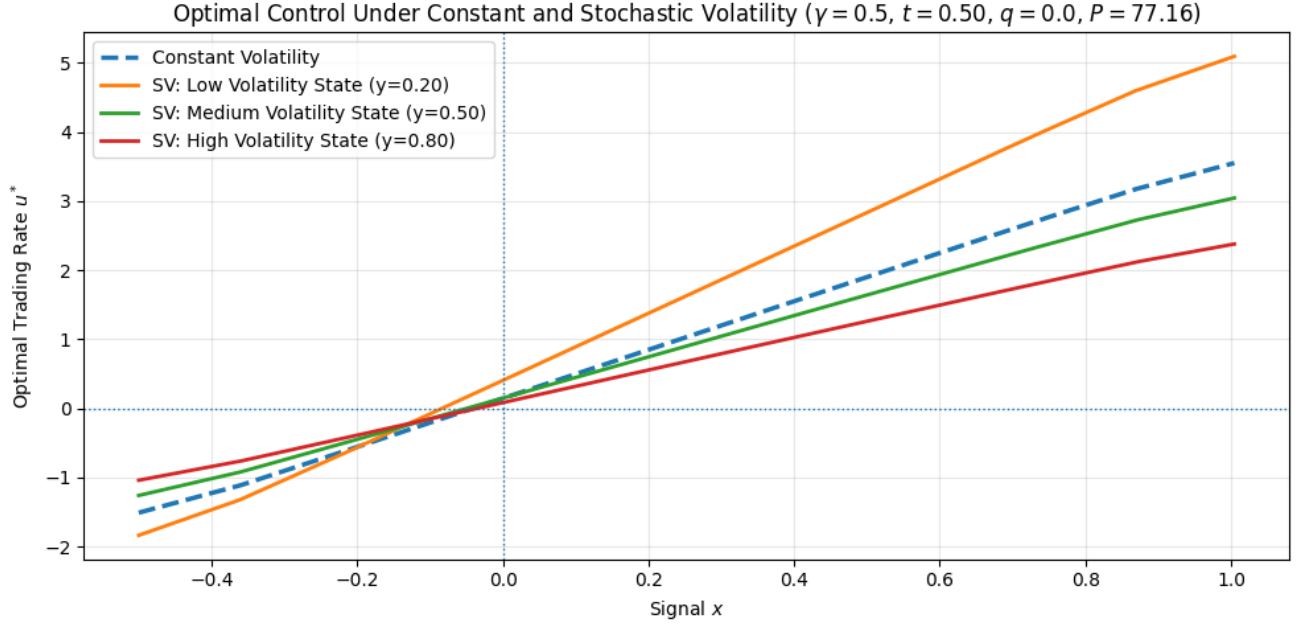


Figure 4.2: Optimal trading rate u^* as a function of the predictive signal under constant and stochastic volatility with $\gamma = 0.5$, evaluated at $t = 0.5$, $q = 0$, and $P = 77.16$.

Figure 4.2 illustrates how stochastic volatility changes the optimal control for a specific state. For each model, the optimal trading rate is increasing in the predictive signal x , so more favorable signals lead to more aggressive buying, while unfavorable signals lead to selling. The key difference is that under stochastic volatility the control also depends on the current volatility, so it can adjust u^* based on that state as well. In the low-volatility state, the strategy trades more aggressively than the constant-volatility benchmark for any given signal. In the high-volatility state, the strategy trades less aggressively. Thus, stochastic volatility changes the slope of the policy function, meaning that the trader scales up exposure when volatility is low and scales it back when volatility is high. This provides a direct policy-level explanation for the earlier sample-path figure. The inventory there evolves smoothly not only because of transaction costs and risk aversion, but also because the stochastic-volatility strategy continuously adjusts its responsiveness to the signal depending on the prevailing volatility regime.

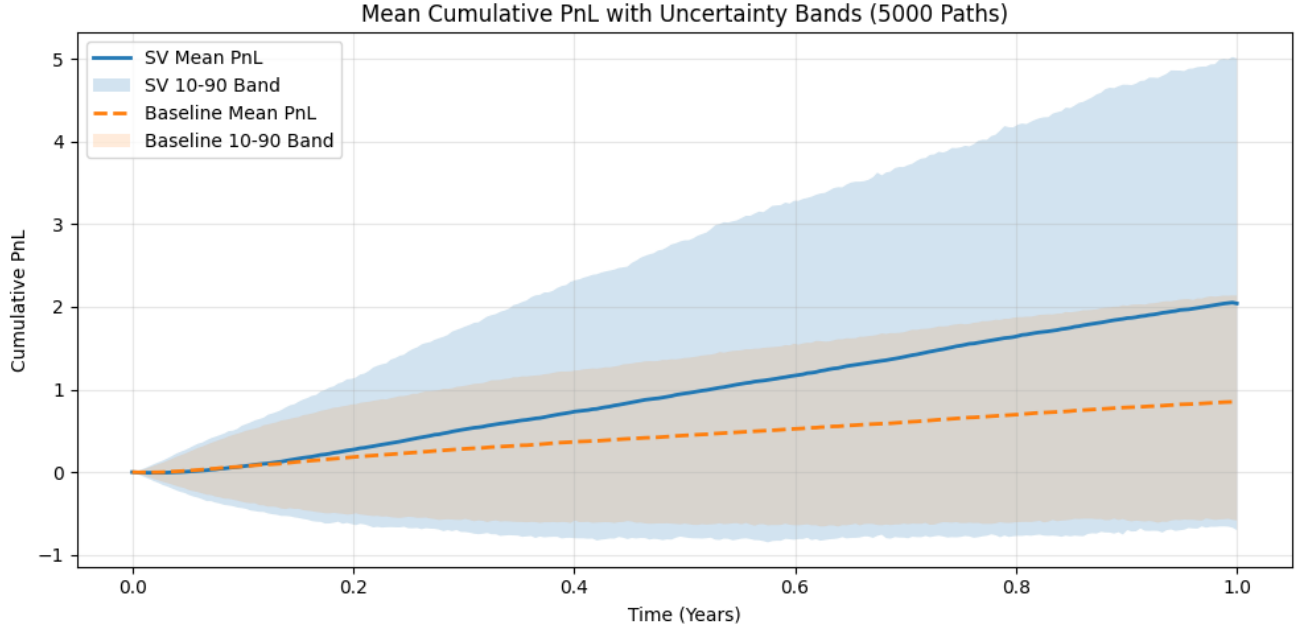


Figure 4.3: Mean cumulative PnL with 10–90 percentile bands for the stochastic-volatility and constant-volatility strategies.

Figure 4.3 plots the mean cumulative PnL across 5,000 paths for the stochastic-volatility strategy and the constant-volatility baseline with uncertainty bands at $\gamma = 0.5$. The stochastic-volatility strategy produces a higher mean cumulative PnL over the horizon, and the gap becomes larger as time goes on. By the terminal date, the mean cumulative PnL under stochastic volatility is around 2.04, compared with about 0.85 for the baseline. This suggests that allowing the strategy to respond to changing volatility improves performance in expectation. At the same time, the uncertainty band for the stochastic-volatility strategy is much wider. Thus, the gain in average performance comes with greater dispersion in outcomes.

This pattern is reinforced by Figure 4.4, which shows the distribution of terminal PnL across the 5,000 Monte Carlo paths at $\gamma = 0.5$. The stochastic-volatility strategy has mean terminal PnL of about 2.04 with standard deviation about 2.30, while the constant-volatility strategy has mean terminal PnL of about 0.85 with standard deviation about 1.08. The constant-volatility strategy has a much tighter distribution concentrated around modest gains. By contrast, the stochastic-volatility strategy has a much wider distribution and a substantially heavier right tail.

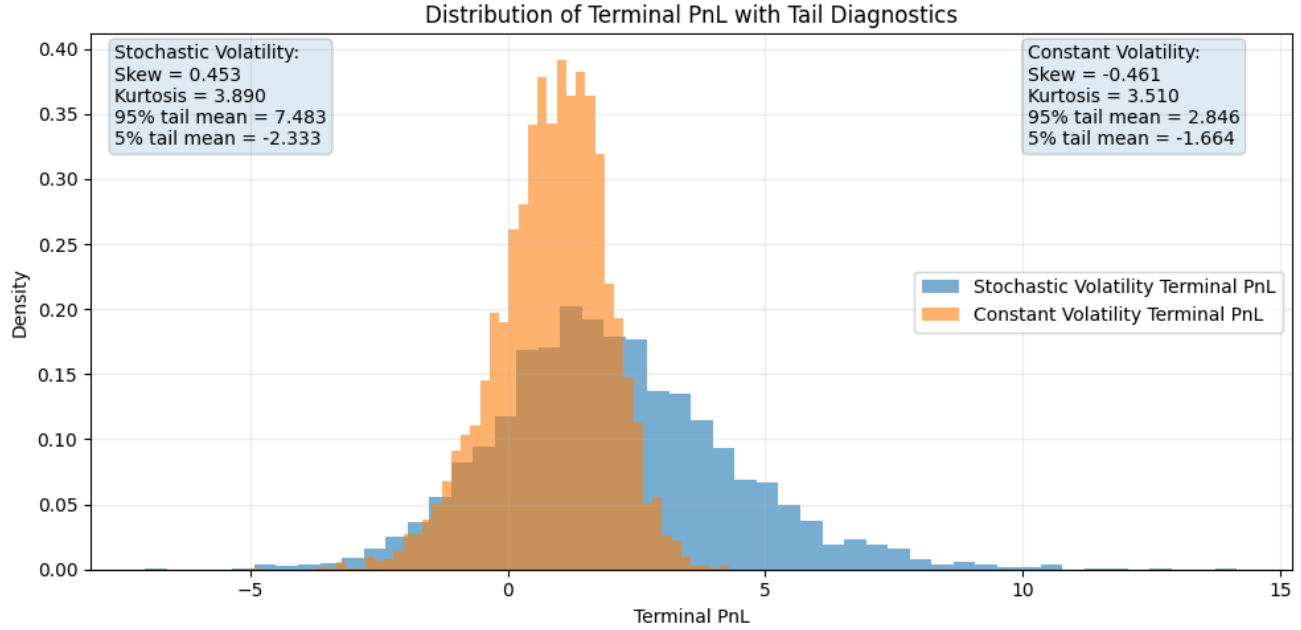


Figure 4.4: Distribution of terminal PnL across 5000 Monte Carlo paths for the stochastic-volatility and constant-volatility strategies along with statistics for the higher-order moments of the distribution.

This indicates that the additional variability in results for the stochastic-volatility strategy is tilted toward large positive outcomes rather than being driven equally by upside and downside realizations.

The asymmetry is also reflected in the higher-order moments of the distribution. The stochastic-volatility strategy has positive skewness, about 0.45, while the constant-volatility strategy has negative skewness, about -0.46. Similarly, the kurtosis of the stochastic-volatility distribution is about 3.89, compared with about 3.51 for the constant-volatility benchmark, indicating heavier tails overall. The upper 5% tail mean is also far larger for the stochastic-volatility strategy than for the constant-volatility strategy, while the lower 5% tail mean is only moderately more negative. Taken together, these results show that although the stochastic-volatility strategy is more volatile overall, a substantial portion of that extra dispersion comes from favorable right-tail outcomes.

Finally, Figure 4.5 compares mean terminal PnL across 2000 paths for several values of the risk-aversion parameter γ . As expected, mean terminal PnL decreases as γ increases for both

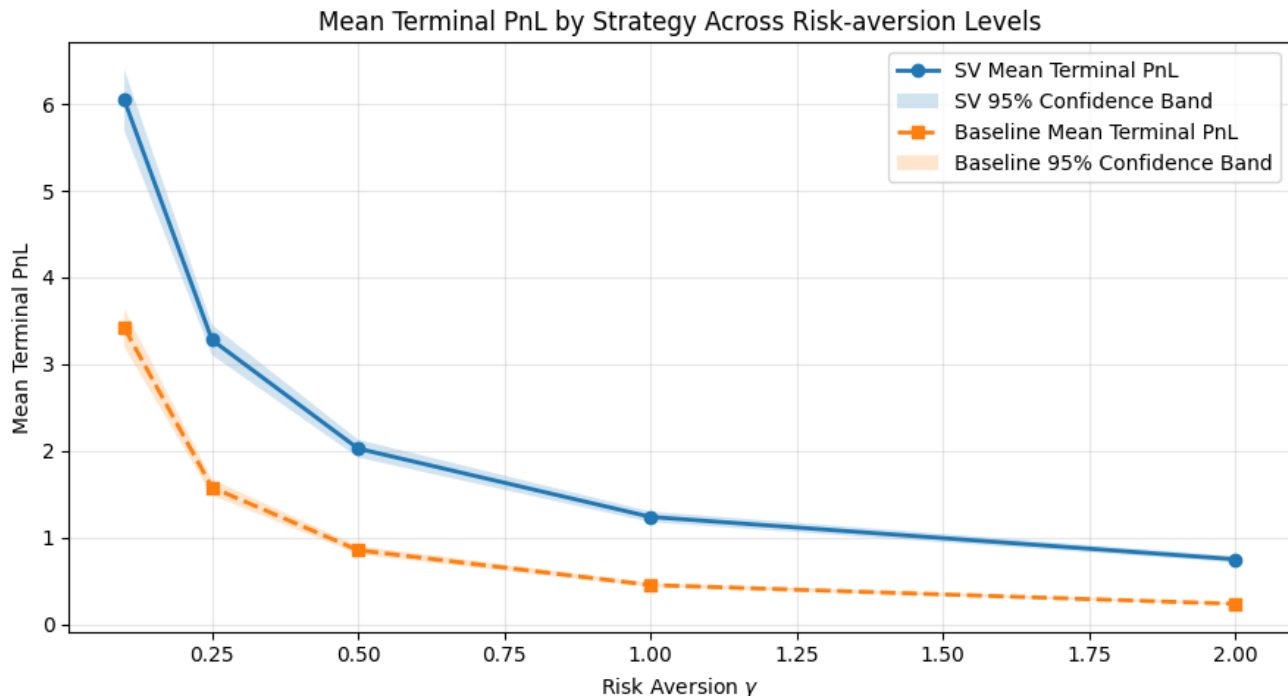


Figure 4.5: Mean terminal PnL by strategy across risk-aversion levels, with 95% confidence bands.

strategies. A more risk-averse trader holds smaller positions and therefore earns lower expected returns. The stochastic-volatility strategy continues to outperform the constant-volatility baseline at every level of γ considered in expectation. The advantage is largest when γ is small and becomes smaller as γ increases because when the trader is already taking only a very small position, there is less benefit from adapting the strategy to the current volatility state.

Overall, the numerical results support the same qualitative conclusion throughout. Incorporating stochastic volatility into the trading rule improves expected performance relative to a constant-volatility benchmark, but it also increases the variability of outcomes with a right skew. Thus, the stochastic-volatility model does not simply produce a higher value function in theory, it also changes the actual shape of the strategy in simulation. In the commodity setting considered here, the ability to respond to changes in volatility appears to be most useful when risk aversion is moderate rather than very high.

4.2.4 Limitations

There are two main limitations to these numerical results. First, the implementation uses a reduced one-factor stochastic-volatility model rather than the full multiscale specification introduced earlier. Second, the numerical grid is kept fairly coarse in order to make the nonlinear backward solve computationally feasible. Because of this, the results should be interpreted mainly as evidence for the qualitative behavior of the strategy rather than as a fully optimized calibration. Still, the solver is sufficient to show that the main conclusions of the model remain present in the commodity setting: the optimal strategy reacts gradually to signals, stochastic volatility improves expected performance, and this improvement comes with a wider distribution of outcomes.

Chapter 5

Conclusion

In this thesis, I adapted the optimal trading framework of Chan et al. to a commodity setting. While their model is developed for equities with predictable returns, transaction costs, price impact, and multiscale stochastic volatility, here I instead modeled the commodity price dynamics through the Schwartz two-factor framework. This modification changes the price process from the additive form in the original paper to a geometric one, which in turn makes the associated Hamilton–Jacobi–Bellman equation nonlinear and removes tractability. Because of this, rather than pursuing a purely asymptotic analysis, I studied the problem through a numerical finite-difference scheme together with Monte Carlo simulation.

The mathematical contribution of this thesis is therefore to formulate the commodity trading problem as a stochastic optimal control problem, derive the corresponding HJB equation, reduce it through a quadratic ansatz in the inventory variable, and solve the resulting coefficient system numerically. In the implementation, I further reduced the stochastic-volatility specification to a one-factor model, since data identify a single volatility factor through OVX. This still preserves the key economic feature of time-varying volatility while keeping the state space manageable enough for the numerical scheme.

The numerical results suggest that incorporating stochastic volatility materially changes the optimal trading strategy. Relative to the constant-volatility benchmark, the stochastic-volatility

strategy produces higher expected PnL, but also a wider distribution of outcomes. In particular, at the benchmark level $\gamma = 0.5$, the stochastic-volatility strategy attains mean terminal PnL of about 2.04, compared with about 0.85 for the constant-volatility baseline. Although the stochastic volatility has a larger dispersion of results and slightly larger left tail, the terminal PnL is positively skewed towards a larger right tail. Across different values of the risk-aversion parameter, the same pattern remains: the stochastic-volatility strategy consistently outperforms in expectation, but the size of this advantage declines as the trader becomes more risk-averse. Thus, the numerical evidence supports the conclusion that volatility-sensitive trading improves average performance, while at the same time increasing variance in the form of a broader range of realized outcomes.

There are several natural directions in which this work could be extended:

- **Modeling futures prices directly.** The price dynamics could be modified to follow futures prices directly rather than the spot commodity. This would better reflect how most financial participants actually trade commodities, and it would avoid treating the spot price as a directly tradable asset when storage and delivery frictions make that unrealistic.
- **Reintroducing persistent price impact.** One could reintroduce persistent price impact into the commodity framework. The present thesis drops this feature for simplicity, but it is an important part of the original Chan et al. model and would make the resulting strategy more realistic for larger traders or less liquid contracts.
- **Using a fully calibrated stochastic-volatility model.** One could incorporate a fully calibrated stochastic-volatility model using high-frequency data and latent volatility states. The current implementation uses a reduced one-factor specification identified from OVX, but a richer calibration procedure could recover both fast and slow volatility components and bring the model closer to the multiscale setting that originally motivated the problem.

Each of these extensions would make the model more realistic, though they would also increase the analytical and numerical difficulty of the problem.

Overall, this thesis shows that the stochastic optimal control framework developed for equity trading can be meaningfully adapted to commodity markets, but that doing so introduces new nonlinearities that make numerical methods necessary. Even in the simplified one-factor setting studied here, stochastic volatility has a clear effect on both the value function and the resulting trading rule. This suggests that extending optimal trading models from equities to commodities is not only mathematically interesting, but also important for understanding how realistic market features affect trading decisions in practice.

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